

SAPIENZA
UNIVERSITÀ DI ROMA

Algorithms

Volume 3 — Asymptotic growth and notation

How the functions of a natural variable that count the steps of an algorithm grow: the families of increasing functions (logarithms, powers, exponentials, factorial), the hierarchy of infinities and the comparison of orders of growth; the asymptotic notations Theta, big O, Omega, little o and omega, their properties and their use in the analysis of algorithms. Limits and functions of a natural variable are taken from the course of Mathematical Analysis 1.

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Chapter 9

Asymptotic growth

Introduction

The number of steps executed by an algorithm is a function of a natural number: the size of its input. Two algorithms that solve the same problem are compared by looking at how fast these functions grow when the input grows, and not at their values on a few small inputs. In this chapter we study the functions of a natural variable that appear in the analysis of algorithms and we compare their **orders of growth**.

The plan of the chapter is the following.

- In Section 9.1 we introduce the five families of increasing functions of a natural variable used in the course: logarithmic, power, exponential, factorial and self-power functions.
- In Section 9.2 we recall the four limits that put these families in order (the **hierarchy of infinities**).
- In Section 9.3 we define the comparison of two orders of growth through the limit of the ratio of the two functions. We then derive the hierarchy of the orders of growth, a classification of the orders of growth, and the properties of logarithmic, power, polynomial and exponential functions that are used in the analysis of algorithms.

Prerequisites. The functions of a natural variable are the **sequences** of the course *Mathematical Analysis 1* (`https://fabiofurini.github.io/mathematical-analysis-1/`, Italian version `https://fabiofurini.github.io/analisi-matematica-1/`). Their limits, and the theorems used to compute them, are not explained again in this chapter: we use them, and we refer to the part *Limits of sequences* of that course, in particular to the chapters *Computing limits of sequences* (comparison theorem), *Euler's number* and *Hierarchies of infinities and the ratio test*.

9.1 Families of increasing functions of a natural variable

A function of a natural variable is defined on a set of natural numbers of the form $\mathbb{N}_{\geq n_0} = \{n_0, n_0+1, n_0+2, \dots\}$, and in the analysis of algorithms only its behaviour for large values of n matters. The following two definitions describe properties that hold *from some value of n onwards*: in the course *Mathematical Analysis 1* they are called properties that hold *eventually*.

Definition 9.1: asymptotically positive function

A function of a natural variable $f : \mathbb{N}_{\geq n_0} \to \mathbb{R}$ is **asymptotically positive** if there exists $n_1 \in \mathbb{N}_{\geq n_0}$ such that:

$$f(n) > 0, \quad \forall n \in \mathbb{N}_{\geq n_1} \tag{9.1}$$

Definition 9.2: asymptotically increasing function

A function of a natural variable $f : \mathbb{N}_{\geq n_0} \to \mathbb{R}$ is **asymptotically increasing** if there exists $n_1 \in \mathbb{N}_{\geq n_0}$ such that:

$$f(n) < f(n+1), \quad \forall n \in \mathbb{N}_{\geq n_1} \tag{9.2}$$

- The principal families of asymptotically increasing functions of a natural variable are the following ones.

  1. The **logarithmic** functions of a natural variable:

  $$f : \mathbb{N}_{\geq 1} \to \mathbb{R}, \quad f : n \mapsto \log_a n \quad \forall a \in \mathbb{R}_{>1} \tag{9.3}$$

  2. The **power** functions of a natural variable:

  $$f : \mathbb{N} \to \mathbb{R}, \quad f : n \mapsto n^\alpha \quad \forall \alpha \in \mathbb{R}_{>0} \tag{9.4}$$

  3. The **exponential** functions of a natural variable:

  $$f : \mathbb{N} \to \mathbb{R}, \quad f : n \mapsto b^n \quad \forall b \in \mathbb{R}_{>1} \tag{9.5}$$

  4. The **factorial** function of a natural variable:

  $$f : \mathbb{N} \to \mathbb{R}, \quad f : n \mapsto n! \tag{9.6}$$

  5. The **self-power** function of a natural variable:

  $$f : \mathbb{N} \to \mathbb{R}, \quad f : n \mapsto n^n \tag{9.7}$$

  where, by convention, we set $0^0 = 1$.

- All these functions are asymptotically positive, and all of them diverge to $+\infty$ as $n \to +\infty$. The factorial and the self-power functions are not increasing on their whole domain, since $0! = 1! = 1$ and $0^0 = 1^1 = 1$, but they are strictly increasing from $n = 1$ onwards: this is why we speak of *asymptotically* increasing functions.

- In these notes $\ln n = \log_e n$ denotes the natural logarithm, where e is Euler's number.

Example 9.1: increasing functions of a natural variable

The table reports the values of some increasing functions of a natural variable, computed by increasing n by one order of magnitude at a time, from 1 to 100 million. The values of $\ln n$ and the leading factors of 2^n and $n!$ are rounded to one decimal digit; the values of n^2 and n^n are exact.

n	$\ln n$	n^2	2^n	$n!$	n^n
$1 = 10^0$	0	1	2	1	1
$10 = 10^1$	≈ 2.3	10^2	$\approx 1.0 \cdot 10^3$	$\approx 3.6 \cdot 10^6$	10^{10}
$100 = 10^2$	≈ 4.6	10^4	$\approx 1.3 \cdot 10^{30}$	$\approx 9.3 \cdot 10^{157}$	10^{200}
$1{,}000 = 10^3$	≈ 6.9	10^6	$\approx 1.1 \cdot 10^{301}$	$\approx 4.0 \cdot 10^{2{,}567}$	$10^{3{,}000}$
$10{,}000 = 10^4$	≈ 9.2	10^8	$\approx 2.0 \cdot 10^{3{,}010}$	$\approx 2.8 \cdot 10^{35{,}659}$	$10^{40{,}000}$
$100{,}000 = 10^5$	≈ 11.5	10^{10}	$\approx 1.0 \cdot 10^{30{,}103}$	$\approx 2.8 \cdot 10^{456{,}573}$	$10^{500{,}000}$
10^6	≈ 13.8	10^{12}	$\approx 9.9 \cdot 10^{301{,}029}$	$\approx 8.3 \cdot 10^{5{,}565{,}708}$	$10^{6{,}000{,}000}$
10^7	≈ 16.1	10^{14}	$\approx 9.0 \cdot 10^{3{,}010{,}299}$	$\approx 1.2 \cdot 10^{65{,}657{,}059}$	$10^{70{,}000{,}000}$
10^8	≈ 18.4	10^{16}	$\approx 3.7 \cdot 10^{30{,}102{,}999}$	$\approx 1.6 \cdot 10^{756{,}570{,}556}$	$10^{800{,}000{,}000}$

To appreciate these numbers, recall that the number of atoms in the observable universe is estimated at about 10^{82}, and that a personal computer of 2024 executes about $10 \cdot 10^9 = 10^{10}$ operations per second (10 GFLOPS). An algorithm that executes 2^n operations on an input with $n = 100$ needs

$$\frac{2^{100}}{10^{10}} \text{ seconds} \approx 1.3 \cdot 10^{20} \text{ seconds} \approx 4.0 \cdot 10^{12} \text{ years},$$

while an algorithm that executes n^2 operations needs $10^4/10^{10} = 10^{-6}$ seconds.

- Figure 9.1 shows the first values of the same functions. Since the values grow very fast, the vertical axis is in logarithmic scale: each horizontal line of the grid corresponds to a multiplication by 100.

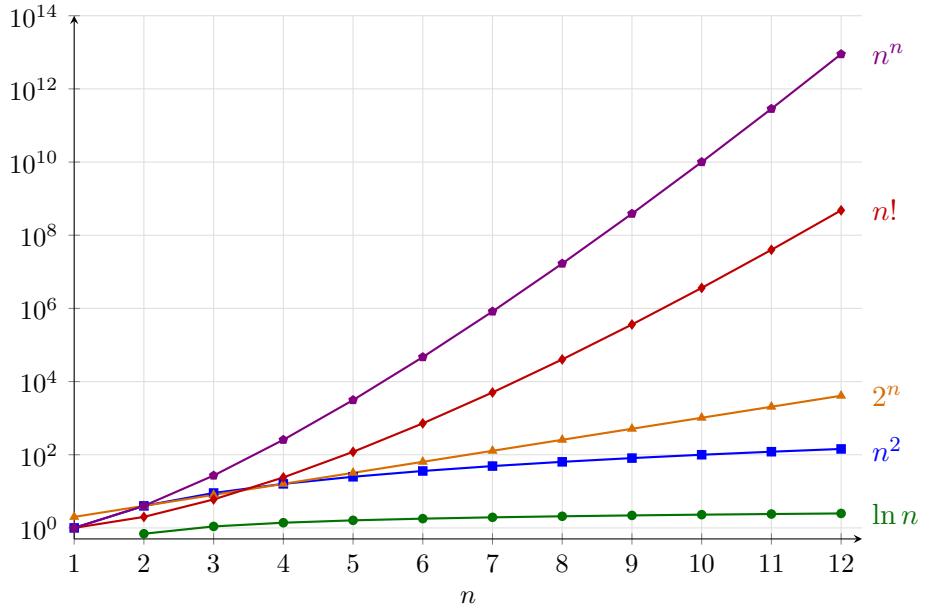


Figure 9.1: The values of $\ln n$, n^2, 2^n, $n!$ and n^n for $n \in \{1, 2, \ldots, 12\}$ (vertical axis in logarithmic scale; the points are joined by segments only for readability). For small values of n the order of the functions can be different from the final one: $2^3 = 8 < 9 = 3^2$, and $2^n > n!$ for $n \in \{1, 2, 3\}$.

9.2 The hierarchy of infinities

The five families of Section 9.1 all diverge to $+\infty$. To decide which one diverges faster, we compute the limit of the ratio of two of them. The limits of this section are the *hierarchy of infinities* for sequences of the course *Mathematical Analysis 1* (chapter *Hierarchies of infinities and the ratio test*), where they are proved; here we recall them, together with the idea of their proofs, and we extend the first one to the powers of the logarithms.

- Given $a \in \mathbb{R}_{>1}$ and $\beta \in \mathbb{R}_{>0}$, the asymptotically increasing function of a natural variable

$$f : \mathbb{N}_{\geq 1} \to \mathbb{R}, \quad f : n \mapsto \log_a^\beta n = \left(\log_a n\right)^\beta \tag{9.8}$$

  is called a **polylogarithmic** function of a natural variable.

Proposition 9.3: logarithms and powers

For every $\alpha \in \mathbb{R}_{>0}$, $\beta \in \mathbb{R}_{>0}$ and $a \in \mathbb{R}_{>1}$, we have:

$$\lim_{n \to +\infty} \frac{\log_a^\beta n}{n^\alpha} = 0^+ \tag{9.9}$$

Proof. For $\beta = 1$ this is the hierarchy of infinities (part I) of the course *Mathematical*

Analysis 1:

$$\lim_{n\to+\infty} \frac{\log_a n}{n^\alpha} = 0^+, \quad \forall \alpha \in \mathbb{R}_{>0},\ \forall a \in \mathbb{R}_{>1}.$$

It is proved with the inequality $\log_a x < x \log_a 2$, valid for every $x \in \mathbb{R}_{>0}$ (a consequence of Bernoulli's inequality), applied to $x = n^{\alpha/2}$, and with the comparison theorem.

For every $\beta \in \mathbb{R}_{>0}$, since $\frac{\alpha}{\beta} \in \mathbb{R}_{>0}$, we then have:

$$\lim_{n\to+\infty} \frac{\log_a^\beta n}{n^\alpha} = \lim_{n\to+\infty} \left(\underbrace{\frac{\log_a n}{n^{\alpha/\beta}}}_{\to 0^+} \right)^\beta = 0^+$$

□

Proposition 9.4: powers and exponentials

For every $\alpha \in \mathbb{R}_{>0}$ and $b \in \mathbb{R}_{>1}$, we have:

$$\lim_{n\to+\infty} \frac{n^\alpha}{b^n} = 0^+ \tag{9.10}$$

- This is the hierarchy of infinities (part II). It follows from Proposition 9.3 by replacing n with 2^n, which gives $\frac{n}{(2^\mu)^n} \to 0^+$ for every $\mu \in \mathbb{R}_{>0}$, hence $\frac{n}{b^n} \to 0^+$ for every $b \in \mathbb{R}_{>1}$ (choose μ such that $2^\mu = b$); the general case follows from the identity

$$\frac{n^\alpha}{b^n} = \left(\frac{n}{(b^{1/\alpha})^n} \right)^\alpha, \qquad b^{1/\alpha} \in \mathbb{R}_{>1}.$$

Proposition 9.5: exponentials and factorial

For every $b \in \mathbb{R}_{>1}$, we have:

$$\lim_{n\to+\infty} \frac{b^n}{n!} = 0^+ \tag{9.11}$$

- This is the hierarchy of infinities (part III). It follows from the ratio test, since

$$\frac{b^{n+1}}{(n+1)!}\,\frac{n!}{b^n} = \frac{b}{n+1} \to 0^+ < 1.$$

Proposition 9.6: factorial and self-power

$$\lim_{n\to+\infty} \frac{n!}{n^n} = 0^+ \tag{9.12}$$

- This is the hierarchy of infinities (part IV). It follows from the ratio test, since

$$\frac{(n+1)!}{(n+1)^{n+1}}\,\frac{n^n}{n!} = \left(\frac{n}{n+1} \right)^n = \frac{1}{\left(1+\frac{1}{n}\right)^n} \to \frac{1}{e} < 1.$$

Polylogarithmic functions (with base greater than one) diverge more slowly than power functions (with positive exponent), which diverge more slowly than exponential functions (with base greater than one), which diverge more slowly than the factorial function, which diverges more slowly than the self-power function.

9.3 Comparison of orders of growth

- The **order of growth** of an asymptotically increasing function of a natural variable describes how rapidly its values increase compared to the values of another asymptotically increasing function of a natural variable.
- To compare two functions f and g we look at the limit of their ratio $\frac{f(n)}{g(n)}$, which is defined for all sufficiently large n when f and g are asymptotically positive.

Definition 9.7: comparison of orders of growth

Let f and g be two asymptotically positive functions of a natural variable such that the limit

$$L = \lim_{n \to +\infty} \frac{f(n)}{g(n)}$$

exists (finite or $+\infty$). We say that:

- f has a strictly **smaller order of growth** than g, if $L = 0^+$;
- f has a **smaller or equal order of growth** compared to g, if $L = 0^+$ or $L \in \mathbb{R}_{>0}$;
- f has the **same order of growth** as g, if $L \in \mathbb{R}_{>0}$;
- f has a **greater or equal order of growth** compared to g, if $L \in \mathbb{R}_{>0}$ or $L = +\infty$;
- f has a strictly **greater order of growth** than g, if $L = +\infty$.

Given two asymptotically increasing functions of a natural variable f and g:

- if f has a strictly smaller order of growth than g, $\rightsquigarrow$ the function f **grows slower than** the function g;
- if f has a smaller or equal order of growth compared to g, $\rightsquigarrow$ the function f **grows at most as fast as** the function g;
- if f has the same order of growth as g, $\rightsquigarrow$ the function f **grows as fast as** the function g;
- if f has a greater or equal order of growth compared to g, $\rightsquigarrow$ the function f **grows at least as fast as** the function g;
- if f has a strictly greater order of growth than g, $\rightsquigarrow$ the function f **grows faster than** the function g.

- The comparison does not depend on the values of f and g on finitely many natural numbers: changing them does not change the limit of the ratio.
- The comparison is symmetric, as stated by the following observation.

Observation 9.8: exchanging the two functions

Let f and g be two asymptotically positive functions of a natural variable. Then:

- f has a strictly smaller order of growth than g if and only if g has a strictly greater order of growth than f;
- f has the same order of growth as g if and only if g has the same order of growth as f.

Proof. For all sufficiently large n we have $\frac{g(n)}{f(n)} = 1 \Big/ \frac{f(n)}{g(n)}$, with both ratios positive. By the theorem on the limit of the reciprocal of a sequence:

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = 0^+ \quad \iff \quad \lim_{n\to+\infty} \frac{g(n)}{f(n)} = +\infty$$

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \quad \iff \quad \lim_{n\to+\infty} \frac{g(n)}{f(n)} = \frac{1}{\ell} \in \mathbb{R}_{>0}$$

□

Example 9.2: the limit of the ratio

We compare three pairs of functions; the three ratios are drawn in Figure 9.2.

- The function $n \mapsto \ln n$ has a strictly smaller order of growth than $n \mapsto \sqrt{n} = n^{1/2}$, since by Proposition 9.3 (with $a = e$, $\beta = 1$, $\alpha = \frac{1}{2}$):

$$\lim_{n\to+\infty} \frac{\ln n}{\sqrt{n}} = 0^+$$

- The function $n \mapsto 2n^2 + 3n$ has the same order of growth as $n \mapsto n^2$, since:

$$\lim_{n\to+\infty} \frac{2n^2+3n}{n^2} = \lim_{n\to+\infty} \Big(2 + \underbrace{\frac{3}{n}}_{\to 0}\Big) = 2 \in \mathbb{R}_{>0}$$

- The function $n \mapsto n \ln n$ has a strictly greater order of growth than $n \mapsto n$, since:

$$\lim_{n\to+\infty} \frac{n \ln n}{n} = \lim_{n\to+\infty} \ln n = +\infty$$

The first ratio is not monotone: it increases up to its largest value, about 0.74, reached for $n = 7$, and only then it decreases towards 0.

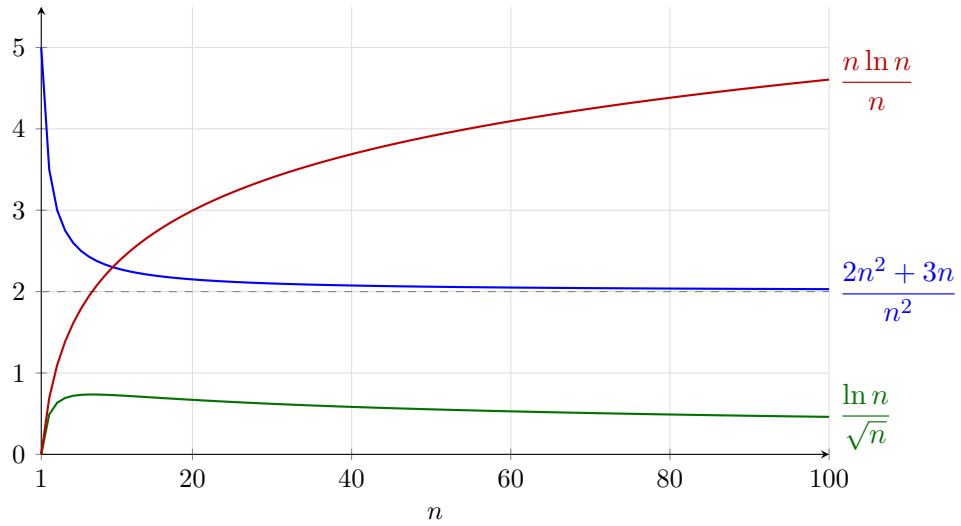


Figure 9.2: The three ratios of Example 9.2 for $n \in \{1, 2, \ldots, 100\}$ (the points are joined by segments). The first one tends to 0, the second one to 2 (dashed line), the third one diverges to $+\infty$.

- The comparison of Definition 9.7 requires the limit of the ratio to exist. For example, for $f(n) = n$ if n is even and $f(n) = 2n$ if n is odd, the ratio $\frac{f(n)}{n}$ takes alternately the values 1 and 2 and has no limit. All the comparisons of this chapter are made between functions whose ratio has a limit.

When two functions have the same order of growth, the ratio of their values stays, for all sufficiently large n, between two positive constants.

Observation 9.9: bounds on the ratio

Let f and g be two asymptotically positive functions of a natural variable with the same order of growth, and let $\ell = \lim_{n\to+\infty} \frac{f(n)}{g(n)} \in \mathbb{R}_{>0}$. Then there exist $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that:

$$0 < c_1\, g(n) \leq f(n) \leq c_2\, g(n), \quad \forall n \in \mathbb{N}_{\geq n_1} \tag{9.13}$$

Proof. By the definition of limit with $\varepsilon = \frac{\ell}{2} > 0$, there exists $n_1 \in \mathbb{N}$, which we can choose so large that $f(n) > 0$ and $g(n) > 0$ for every $n \geq n_1$, such that:

$$\left|\frac{f(n)}{g(n)} - \ell\right| < \frac{\ell}{2} \quad \Longleftrightarrow \quad \frac{\ell}{2} < \frac{f(n)}{g(n)} < \frac{3\,\ell}{2}, \quad \forall n \in \mathbb{N}_{\geq n_1}$$

Multiplying by $g(n) > 0$, the claim follows with $c_1 = \frac{\ell}{2}$ and $c_2 = \frac{3\,\ell}{2}$. □

- Accordingly, if two constants $c_1, c_2 \in \mathbb{R}_{>0}$ as in Observation 9.9 do not exist, then f and g do not have the same order of growth.

9.3.1 Hierarchy of the orders of growth

The limits of Section 9.2 translate directly into comparisons of orders of growth.

- Any polylogarithmic function of a natural variable with a base strictly greater than one has a strictly smaller order of growth than any power function of a natural variable with a positive exponent.

Observation 9.10: polylogarithms and powers

For every $\alpha \in \mathbb{R}_{>0}$, $\beta \in \mathbb{R}_{>0}$ and $a \in \mathbb{R}_{>1}$, the function $n \mapsto \log_a^\beta n$ has a strictly smaller order of growth than the function $n \mapsto n^\alpha$.

Proof. By Proposition 9.3, for every $\alpha \in \mathbb{R}_{>0}$, $\beta \in \mathbb{R}_{>0}$ and $a \in \mathbb{R}_{>1}$, we have:

$$\lim_{n \to +\infty} \frac{\log_a^\beta n}{n^\alpha} = 0^+$$

□

- Any power function of a natural variable with a positive exponent has a strictly smaller order of growth than any exponential function of a natural variable with a base strictly greater than one.

Observation 9.11: powers and exponentials

For every $\alpha \in \mathbb{R}_{>0}$ and $b \in \mathbb{R}_{>1}$, the function $n \mapsto n^\alpha$ has a strictly smaller order of growth than the function $n \mapsto b^n$.

Proof. By Proposition 9.4, for every $\alpha \in \mathbb{R}_{>0}$ and $b \in \mathbb{R}_{>1}$, we have:

$$\lim_{n \to +\infty} \frac{n^\alpha}{b^n} = 0^+$$

□

Example 9.3: comparison of orders of growth

- The function $n \mapsto \ln^6 n + 2 + 2^{1/n} + 3^{-n} + 9\,n^2$ has a strictly smaller order of growth

than the function $n \mapsto n^3$, since:

$$\lim_{n\to+\infty} \frac{\ln^6 n + 2 + 2^{1/n} + 3^{-n} + 9\,n^2}{n^3} = \lim_{n\to+\infty} \left(\underbrace{\frac{\ln^6 n}{n^3}}_{\to 0^+} + \underbrace{\frac{2}{n^3}}_{\to 0^+} + \underbrace{\frac{2^{1/n}}{n^3}}_{\to 0^+} + \underbrace{\frac{3^{-n}}{n^3}}_{\to 0^+} + \underbrace{\frac{9}{n}}_{\to 0^+} \right) = 0^+$$

- The function $n \mapsto 3n^2 + \sqrt{n} + \ln n + 5 + 4^{-n} + 10\,n^{1/3}$ has a strictly smaller order of growth than the function $n \mapsto n^3$, since:

$$\lim_{n\to+\infty} \frac{3n^2 + \sqrt{n} + \ln n + 5 + 4^{-n} + 10\,n^{1/3}}{n^3} = \lim_{n\to+\infty} \left(\underbrace{\frac{3}{n}}_{\to 0^+} + \underbrace{\frac{1}{n^{5/2}}}_{\to 0^+} + \underbrace{\frac{\ln n}{n^3}}_{\to 0^+} + \underbrace{\frac{5}{n^3}}_{\to 0^+} + \underbrace{\frac{4^{-n}}{n^3}}_{\to 0^+} + \underbrace{\frac{10}{n^{8/3}}}_{\to 0^+} \right) = 0^+$$

- The function $n \mapsto n^{1/4} + \ln^2 n + \frac{7}{\sqrt{n}} + e^{-n} + 20\,n^{1/2}$ has a strictly smaller order of growth than the function $n \mapsto n^2$, since:

$$\lim_{n\to+\infty} \frac{n^{1/4} + \ln^2 n + \frac{7}{\sqrt{n}} + e^{-n} + 20\,n^{1/2}}{n^2} = \lim_{n\to+\infty} \left(\underbrace{\frac{1}{n^{7/4}}}_{\to 0^+} + \underbrace{\frac{\ln^2 n}{n^2}}_{\to 0^+} + \underbrace{\frac{7}{n^{5/2}}}_{\to 0^+} + \underbrace{\frac{e^{-n}}{n^2}}_{\to 0^+} + \underbrace{\frac{20}{n^{3/2}}}_{\to 0^+} \right) = 0^+$$

In each case the terms with a logarithm go to 0^+ by Observation 9.10, and the other terms by elementary limits.

- Any exponential function of a natural variable with a base strictly greater than one has a strictly smaller order of growth than the factorial function of a natural variable.

Observation 9.12: exponentials and factorial

For every $b \in \mathbb{R}_{>1}$, the function $n \mapsto b^n$ has a strictly smaller order of growth than the function $n \mapsto n!$.

Proof. By Proposition 9.5, for every $b \in \mathbb{R}_{>1}$, we have:

$$\lim_{n\to+\infty} \frac{b^n}{n!} = 0^+$$

□

- The factorial function of a natural variable has a strictly smaller order of growth than the self-power function of a natural variable.

Observation 9.13: factorial and self-power

The function $n \mapsto n!$ has a strictly smaller order of growth than the function $n \mapsto n^n$.

Proof. By Proposition 9.6, we have:

$$\lim_{n \to +\infty} \frac{n!}{n^n} = 0^+$$

□

Example 9.4: comparison with the self-power function

The function $n \mapsto n! + 2^n + \sqrt{n} + \ln^3 n + n^{n-1} + \frac{1}{n}$ has a strictly smaller order of growth than the function $n \mapsto n^n$, since:

$$\lim_{n \to +\infty} \frac{n! + 2^n + \sqrt{n} + \ln^3 n + n^{n-1} + \frac{1}{n}}{n^n} = \lim_{n \to +\infty} \left(\underbrace{\frac{n!}{n^n}}_{\to 0^+} + \underbrace{\frac{2^n}{n^n}}_{\to 0^+} + \underbrace{\frac{\sqrt{n}}{n^n}}_{\to 0^+} + \underbrace{\frac{\ln^3 n}{n^n}}_{\to 0^+} + \underbrace{\frac{1}{n}}_{\to 0^+} + \underbrace{\frac{1}{n^{n+1}}}_{\to 0^+} \right) = 0^+$$

For example, $\frac{2^n}{n^n} = \frac{2^n}{n!} \frac{n!}{n^n}$ is the product of two functions that tend to 0^+, by Propositions 9.5 and 9.6.

Hierarchy of the orders of growth. For every $a \in \mathbb{R}_{>1}$, $\beta \in \mathbb{R}_{>0}$, $\alpha \in \mathbb{R}_{>0}$ and $b \in \mathbb{R}_{>1}$, each function of the chain

$$\log_a^\beta n \qquad n^\alpha \qquad b^n \qquad n! \qquad n^n$$

has a strictly smaller order of growth than the following one.

9.3.2 Classification of orders of growth

The families of the hierarchy are used as yardsticks to classify the order of growth of any other function.

Definition 9.14: bounded orders of growth

An asymptotically increasing function of a natural variable f has:

- a **polylogarithmically bounded order of growth**, if there exists $\beta \in \mathbb{R}_{>0}$ such that f has a smaller or equal order of growth compared to $n \mapsto \ln^\beta n$; and a **logarithmically bounded order of growth**, if this holds with $\beta = 1$;
- a **polynomially bounded order of growth**, if there exists $\alpha \in \mathbb{R}_{>0}$ such that f has a smaller or equal order of growth compared to $n \mapsto n^\alpha$;

- an **exponentially bounded order of growth**, if there exists $b \in \mathbb{R}_{>1}$ such that f has a smaller or equal order of growth compared to $n \mapsto b^n$;
- a **factorially bounded order of growth**, if f has a smaller or equal order of growth compared to $n \mapsto n!$.

Definition 9.15: linear, sub-linear and super-linear order of growth

An asymptotically increasing function of a natural variable f has:

- a **sub-linear order of growth**, if f has a strictly smaller order of growth than $n \mapsto n$;
- a **linear order of growth**, if f has the same order of growth as $n \mapsto n$;
- a **super-linear order of growth**, if f has a strictly greater order of growth than $n \mapsto n$.

- By the hierarchy of the orders of growth, a function with a polylogarithmically bounded order of growth has also a polynomially bounded order of growth, which in turn is also exponentially bounded and factorially bounded.

Example 9.5: classification of orders of growth

We classify four functions, drawn in Figure 9.3.

- $n \mapsto \sqrt{n}$ has a sub-linear order of growth, since $\frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}} \to 0^+$; it has a polynomially bounded order of growth (with $\alpha = \frac{1}{2}$), but not a polylogarithmically bounded one, since $\frac{\sqrt{n}}{\ln^\beta n} \to +\infty$ for every $\beta \in \mathbb{R}_{>0}$ (Proposition 9.3 and Observation 9.8).
- $n \mapsto n$ has a linear order of growth.
- $n \mapsto n \ln n$ has a super-linear order of growth, since $\frac{n \ln n}{n} = \ln n \to +\infty$; it has a polynomially bounded order of growth, since $\frac{n \ln n}{n^2} = \frac{\ln n}{n} \to 0^+$.
- $n \mapsto n^2$ has a super-linear order of growth, since $\frac{n^2}{n} = n \to +\infty$; it has a polynomially bounded order of growth (with $\alpha = 2$).

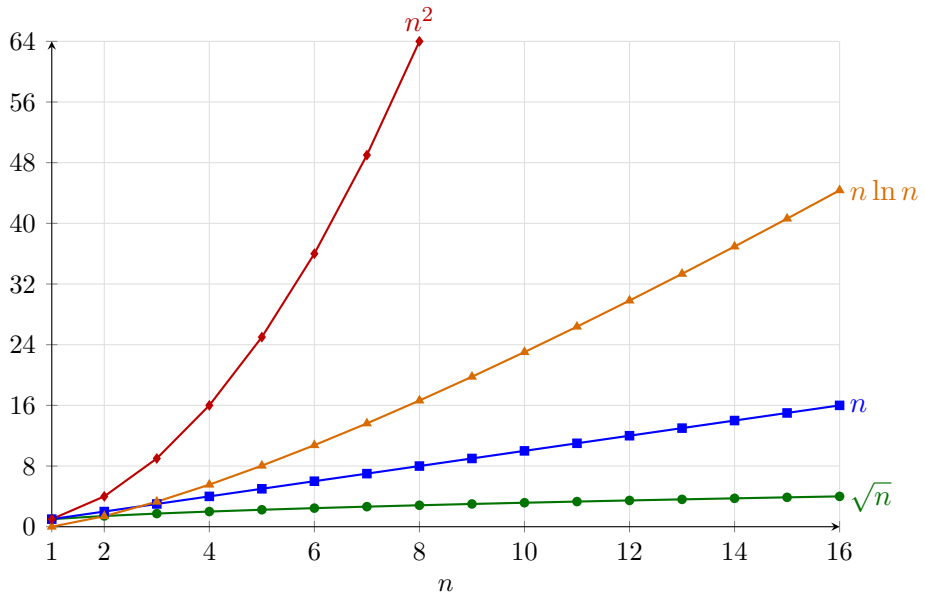


Figure 9.3: A sub-linear function ($\sqrt{n}$), the linear function n and two super-linear functions ($n \ln n$ and n^2) for $n \in \{1, 2, \ldots, 16\}$ (n^2 only up to $n = 8$, where it reaches 64; the points are joined by segments).

9.3.3 Properties of logarithmic functions

- Any logarithmic function of a natural variable with a base strictly greater than one has the same order of growth as any other logarithmic function of a natural variable with a base strictly greater than one; multiplying the argument by a positive constant does not change the order of growth either.

Observation 9.16: change of base

For every $a \in \mathbb{R}_{>1}$, $c \in \mathbb{R}_{>1}$ and $\lambda \in \mathbb{R}_{>0}$, the function $n \mapsto \log_a(\lambda n)$ has the same order of growth as the function $n \mapsto \log_c n$.

Proof. For every $a, c \in \mathbb{R}_{>1}$ and $\lambda \in \mathbb{R}_{>0}$, since $\log_a(\lambda n) = \log_a \lambda + \log_a n$ and $\log_a n = \frac{\log_c n}{\log_c a}$, we have:

$$\lim_{n \to +\infty} \frac{\log_a(\lambda n)}{\log_c n} = \lim_{n \to +\infty} \left(\underbrace{\frac{\log_a \lambda}{\log_c n}}_{\to 0} + \frac{\frac{\log_c n}{\log_c a}}{\log_c n} \right) = \frac{1}{\log_c a} \in \mathbb{R}_{>0}$$

□

- For this reason, when only the order of growth matters, the base of a logarithm is often omitted. For example, $\log_2 n$ and $\ln n$ have the same order of growth, and $\lim_{n \to +\infty} \frac{\log_2 n}{\ln n} = \frac{1}{\ln 2} \approx 1.44$.

- Adding a positive constant to the argument of a logarithmic function of a natural variable does not change its order of growth.

Observation 9.17: shift of the argument

For every $a \in \mathbb{R}_{>1}$ and $\lambda \in \mathbb{R}_{>0}$, the function $n \mapsto \log_a(n+\lambda)$ has the same order of growth as the function $n \mapsto \log_a n$.

Proof. For every $a \in \mathbb{R}_{>1}$ and $\lambda \in \mathbb{R}_{>0}$, we have:

$$\lim_{n\to+\infty} \frac{\log_a(n+\lambda)}{\log_a n} = \lim_{n\to+\infty} \frac{\log_a\left(n\left(1+\frac{\lambda}{n}\right)\right)}{\log_a n} = \lim_{n\to+\infty} \frac{\log_a n + \overbrace{\log_a\left(1+\frac{\lambda}{n}\right)}^{\to 0}}{\log_a n} = 1$$

□

- The **quasi-linear** function of a natural variable $n \mapsto n\log_a n$ has a strictly smaller order of growth than any power function of a natural variable with exponent strictly greater than one.

Observation 9.18: quasi-linear and powers

For every $\alpha \in \mathbb{R}_{>1}$ and $a \in \mathbb{R}_{>1}$, the function $n \mapsto n\log_a n$ has a strictly smaller order of growth than the function $n \mapsto n^\alpha$.

Proof. For every $\alpha \in \mathbb{R}_{>1}$ and $a \in \mathbb{R}_{>1}$, we have $\alpha - 1 \in \mathbb{R}_{>0}$ and, by Proposition 9.3:

$$\lim_{n\to+\infty} \frac{n\log_a n}{n^\alpha} = \lim_{n\to+\infty} \frac{\log_a n}{n^{\alpha-1}} = 0^+$$

□

- The quasi-linear function of a natural variable $n \mapsto n\log_a n$ has a strictly greater order of growth than the linear function of a natural variable: it has a super-linear order of growth.

Observation 9.19: quasi-linear and linear

For every $a \in \mathbb{R}_{>1}$, the function $n \mapsto n\log_a n$ has a strictly greater order of growth than the function $n \mapsto n$.

Proof. For every $a \in \mathbb{R}_{>1}$, we have:

$$\lim_{n \to +\infty} \frac{n \log_a n}{n} = \lim_{n \to +\infty} \log_a n = +\infty$$

□

9.3.4 Properties of power functions

- Adding a positive constant to the argument of a power function of a natural variable does not change its order of growth.

Observation 9.20: shift of the argument

For every $\alpha \in \mathbb{R}_{>0}$ and $\lambda \in \mathbb{R}_{>0}$, the function $n \mapsto (n+\lambda)^\alpha$ has the same order of growth as the function $n \mapsto n^\alpha$.

Proof. For every $\alpha \in \mathbb{R}_{>0}$ and $\lambda \in \mathbb{R}_{>0}$, we have:

$$\lim_{n \to +\infty} \frac{(n+\lambda)^\alpha}{n^\alpha} = \lim_{n \to +\infty} \left(1 + \underbrace{\frac{\lambda}{n}}_{\to 0}\right)^\alpha = 1$$

□

- Multiplying the argument of a power function of a natural variable by a positive constant does not change its order of growth.

Observation 9.21: scaling of the argument

For every $\alpha \in \mathbb{R}_{>0}$ and $\lambda \in \mathbb{R}_{>0}$, the function $n \mapsto (\lambda n)^\alpha$ has the same order of growth as the function $n \mapsto n^\alpha$.

Proof. For every $\alpha \in \mathbb{R}_{>0}$ and $\lambda \in \mathbb{R}_{>0}$, we have:

$$\lim_{n \to +\infty} \frac{(\lambda n)^\alpha}{n^\alpha} = \lim_{n \to +\infty} \frac{\lambda^\alpha n^\alpha}{n^\alpha} = \lambda^\alpha \in \mathbb{R}_{>0}$$

□

- The order of growth of a power function of a natural variable, compared to the linear function of a natural variable, depends on whether the exponent is greater or smaller than one.

Observation 9.22: powers and the linear function

For every $\alpha \in \mathbb{R}_{>0}$, the function $n \mapsto n^\alpha$ has:

- a strictly greater order of growth than the function $n \mapsto n$ (super-linear order of growth), if $\alpha > 1$;
- a strictly smaller order of growth than the function $n \mapsto n$ (sub-linear order of growth), if $\alpha < 1$.

Proof. For every $\alpha \in \mathbb{R}_{>0}$, we have:

$$\lim_{n \to +\infty} \frac{n^\alpha}{n} = \lim_{n \to +\infty} n^{\alpha - 1} = \begin{cases} +\infty & \text{if} \quad \alpha > 1 \\ 0^+ & \text{if} \quad \alpha < 1 \end{cases}$$

□

9.3.5 Properties of polynomial functions

- Any polynomial function of a natural variable with a positive leading coefficient has the same order of growth as its leading term.

Observation 9.23: polynomials

For every $k \in \mathbb{N}_{\geq 1}$ and every $a_0, a_1, \ldots, a_k \in \mathbb{R}$ with $a_k \in \mathbb{R}_{>0}$, the function $n \mapsto \sum_{i=0}^{k} a_i \, n^i$ has the same order of growth as the function $n \mapsto n^k$.

Proof. We have:

$$\lim_{n \to +\infty} \frac{\sum_{i=0}^{k} a_i \, n^i}{n^k} = a_k + \lim_{n \to +\infty} \sum_{i=0}^{k-1} \underbrace{\frac{a_i}{n^{k-i}}}_{\to 0} = a_k \in \mathbb{R}_{>0}$$

In particular, the polynomial is asymptotically positive, so the comparison of Definition 9.7 applies. □

9.3.6 Properties of exponential functions

- Any exponential function of a natural variable with a base strictly greater than one has a strictly greater order of growth than the exponential functions of a natural variable with a smaller base, and a strictly smaller order of growth than the exponential functions of a natural variable with a greater base.

Observation 9.24: different bases

For every $b \in \mathbb{R}_{>1}$ and $c \in \mathbb{R}_{>1}$, the function $n \mapsto b^n$ has:

- a strictly greater order of growth than the function $n \mapsto c^n$, if $b > c$;
- a strictly smaller order of growth than the function $n \mapsto c^n$, if $b < c$.

Proof. For every $b \in \mathbb{R}_{>1}$ and $c \in \mathbb{R}_{>1}$, since $b = c^{\log_c b}$, we have:

$$\lim_{n \to +\infty} \frac{b^n}{c^n} = \lim_{n \to +\infty} \frac{c^{n \log_c b}}{c^n} = \lim_{n \to +\infty} \left(c^{\log_c b - 1}\right)^n = \begin{cases} +\infty & \text{if} \quad b > c \\ 0^+ & \text{if} \quad b < c \end{cases}$$

since $c^{\log_c b - 1} = \frac{b}{c}$ is greater than one if $b > c$ and belongs to $(0, 1)$ if $b < c$ (limit of the geometric progression). □

- Multiplying the exponent of an exponential function of a natural variable by a positive constant changes its order of growth, unless the constant is equal to one.

Observation 9.25: scaling of the exponent

For every $b \in \mathbb{R}_{>1}$ and $\lambda \in \mathbb{R}_{>0}$, the function $n \mapsto b^{\lambda n}$ has:

- a strictly greater order of growth than the function $n \mapsto b^n$, if $\lambda > 1$;
- a strictly smaller order of growth than the function $n \mapsto b^n$, if $0 < \lambda < 1$.

Proof. For every $b \in \mathbb{R}_{>1}$ and $\lambda \in \mathbb{R}_{>0}$, we have:

$$\lim_{n \to +\infty} \frac{b^{\lambda n}}{b^n} = \lim_{n \to +\infty} \left(\frac{b^\lambda}{b}\right)^n = \lim_{n \to +\infty} \left(b^{\lambda - 1}\right)^n = \begin{cases} +\infty & \text{if} \quad \lambda > 1 \\ 0^+ & \text{if} \quad 0 < \lambda < 1 \end{cases}$$

□

- An alternative proof is based on the fact that

$$b^{\lambda n} = \left(b^\lambda\right)^n \qquad \text{and} \qquad b^\lambda \begin{cases} > b & \text{if} \quad \lambda > 1 \\ < b & \text{if} \quad 0 < \lambda < 1 \end{cases}$$

Accordingly, by Observation 9.24 applied to the bases b^λ and b, the function $n \mapsto b^{\lambda n}$ has a strictly greater order of growth than $n \mapsto b^n$ if $\lambda > 1$, and a strictly smaller one if $0 < \lambda < 1$.

Example 9.6: scaling the exponent

- The function $n \mapsto 2^{n/2}$ has a strictly smaller order of growth than the function $n \mapsto 2^n$, since
$$2^{n/2} = \left(2^{1/2}\right)^n \qquad \text{and} \qquad 2^{1/2} = \sqrt{2} < 2.$$
- The function $n \mapsto 2^{3n/2}$ has a strictly greater order of growth than the function $n \mapsto 2^n$, since
$$2^{3n/2} = \left(2^{3/2}\right)^n \qquad \text{and} \qquad 2^{3/2} = 2\sqrt{2} > 2.$$

- Shifting the exponent of an exponential function of a natural variable by a constant does not change its order of growth.

Observation 9.26: shift of the exponent

For every $b \in \mathbb{R}_{>1}$ and $\lambda \in \mathbb{R}$, the function $n \mapsto b^{n+\lambda}$ has the same order of growth as the function $n \mapsto b^n$.

Proof. For every $b \in \mathbb{R}_{>1}$ and $\lambda \in \mathbb{R}$, we have:
$$\lim_{n\to+\infty} \frac{b^{n+\lambda}}{b^n} = \lim_{n\to+\infty} \frac{b^n\, b^\lambda}{b^n} = b^\lambda \in \mathbb{R}_{>0}$$
□

- For example, $2^{n+10} = 1024 \cdot 2^n$ has the same order of growth as 2^n, while $2^{2n} = 4^n$ has a strictly greater order of growth: a constant added to the exponent is a constant factor, a constant factor in the exponent changes the base.

Key points of the chapter

- The functions of a natural variable used in the analysis of algorithms are logarithmic ($\log_a n$), power (n^α), exponential (b^n), factorial ($n!$) and self-power (n^n) functions, together with their sums and products.
- The order of growth of f is compared with that of g through the limit of $\frac{f(n)}{g(n)}$: 0^+ (strictly smaller), a positive real number (same order of growth), $+\infty$ (strictly greater).
- Hierarchy: $\log_a^\beta n$, n^α, b^n, $n!$, n^n (with $a, b > 1$ and $\alpha, \beta > 0$), each with a strictly

smaller order of growth than the next one.

- The base of a logarithm, a constant added to the argument of a logarithm or of a power, a constant factor in the argument of a power, and a constant added to the exponent of an exponential do not change the order of growth; a polynomial has the order of growth of its leading term.
- A constant factor in the exponent of an exponential, and a different base of an exponential, do change the order of growth.
- A function has a sub-linear, linear or super-linear order of growth if it grows slower than, as fast as, or faster than n; for example $\sqrt{n}$, $5n + 3$ and $n \ln n$.

Chapter 10

Asymptotic notation

Introduction

In the chapter «Asymptotic growth» we met the main families of increasing functions of a natural variable (logarithmic, power, exponential and factorial functions) and we compared how fast they grow by means of limits. In this chapter we give the precise language used to express these comparisons: the **asymptotic notations** Θ, $\mathcal{O}$, Ω, o and ω.

The functions of a natural variable (sequences) and their limits are studied in the course of Mathematical Analysis 1, and we do not repeat them here: we only use the definition of limit and the elementary rules to compute limits.

The plan of the chapter is the following.

- In Section 10.1 we define the notations Θ, $\mathcal{O}$ and Ω, we interpret them graphically and we prove that Θ is the same as $\mathcal{O}$ and Ω together. We also study the sum and the maximum of two functions.
- In Section 10.2 we define the notations o and ω, which express bounds that are **not** asymptotically tight.
- In Section 10.3 we link the five notations to the limit of the ratio $\frac{f(n)}{g(n)}$. This gives a fast way to compare two functions, but it does not always work, since the limit may not exist.
- In Section 10.4 we prove the relational properties of the notations (transitivity, reflexivity, symmetry and transpose symmetry), which make the asymptotic comparison of two functions similar to the comparison of two real numbers, with one important difference.
- In Sections 10.5 and 10.6 we connect the notation with the orders of growth of the chapter «Asymptotic growth» and with the analysis of algorithms of the chapters «Input size» and «Running time».

Every constant (c, c_1, c_2, n_1) used in the examples of the chapter has been checked numerically with a script, for all the values of n from the threshold n_1 up to at least $2 \cdot 10^5$.

Throughout the chapter we use the following terminology.

Definition 10.1: asymptotically non-negative and asymptotically positive functions

A function f of a natural variable is **asymptotically non-negative** if there exists $\tilde{n} \in \mathbb{N}$ such that $f(n) \geq 0$ for all $n \geq \tilde{n}$. It is **asymptotically positive** if there exists $\tilde{n} \in \mathbb{N}$ such that $f(n) > 0$ for all $n \geq \tilde{n}$.

10.1 Θ, $\mathcal{O}$ and Ω notations

10.1.1 Definitions

Definition 10.2: Θ-notation

Given two functions f and g of a natural variable, we write $f(n) = \Theta\big(g(n)\big)$ if there exist $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq c_1\, g(n) \leq f(n) \leq c_2\, g(n), \qquad \forall n \geq n_1. \tag{10.1}$$

- If $f(n) = \Theta\big(g(n)\big)$, then, for sufficiently large values of n, the function f is bounded from above and from below by the function g scaled by two positive constants.
- In this case we say that the function g is an **asymptotic tight bound** for the function f.
- We write $f(n) = \Theta(1)$ to mean that the values $f(n)$ are, for sufficiently large n, bounded from above and from below by two positive constants (here g is the constant function $g(n) = 1$).

Definition 10.3: $\mathcal{O}$-notation

Given two functions f and g of a natural variable, we write $f(n) = \mathcal{O}\big(g(n)\big)$ if there exist $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq f(n) \leq c\, g(n), \qquad \forall n \geq n_1. \tag{10.2}$$

- If $f(n) = \mathcal{O}\big(g(n)\big)$, then, for sufficiently large values of n, the function f is bounded from above by the function g scaled by a positive constant.
- In this case we say that the function g is an **asymptotic upper bound** for the function f.

Definition 10.4: Ω-notation

Given two functions f and g of a natural variable, we write $f(n) = \Omega\big(g(n)\big)$ if there exist $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq c\, g(n) \leq f(n), \qquad \forall n \geq n_1. \tag{10.3}$$

- If $f(n) = \Omega\big(g(n)\big)$, then, for sufficiently large values of n, the function f is bounded from below by the function g scaled by a positive constant.
- In this case we say that the function g is an **asymptotic lower bound** for the function f.

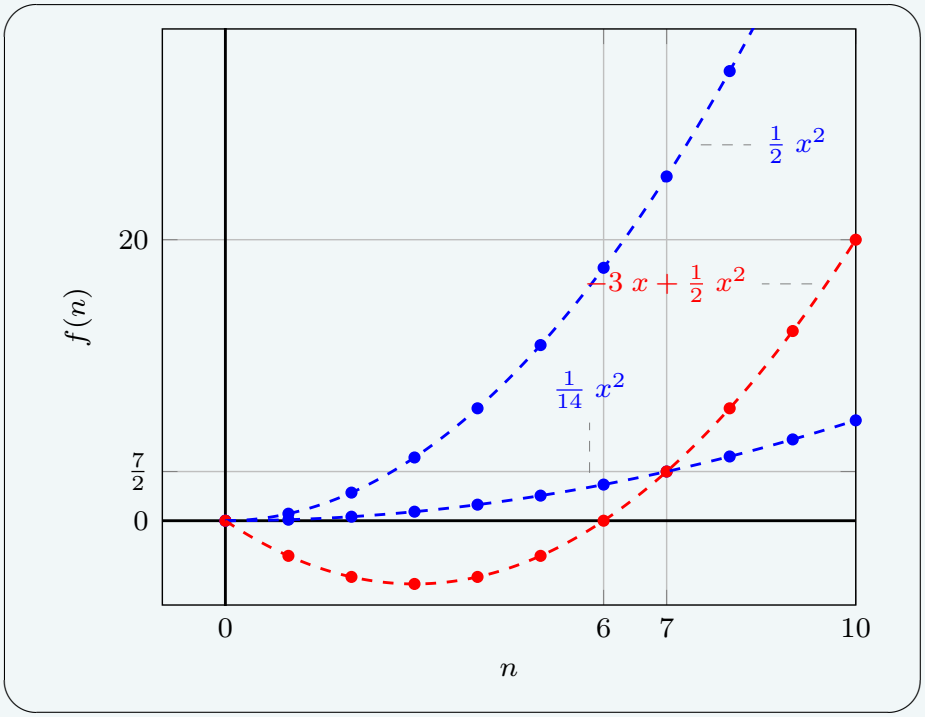
If $f(n) = \Theta\big(g(n)\big)$, $f(n) = \mathcal{O}\big(g(n)\big)$ or $f(n) = \Omega\big(g(n)\big)$ then, since the constants c_1 and c are positive, we have $g(n) \geq 0$ and $f(n) \geq 0$ for all $n \geq n_1$ (for $\mathcal{O}$, the inequality $0 \leq f(n) \leq c\,g(n)$ gives $g(n) \geq 0$ as well). Therefore, the two functions f and g are **asymptotically non-negative**.

- To prove one of these statements we **determine** the constants (c, or c_1 and c_2) and the threshold n_1. In general, the threshold n_1 is not the smallest possible one and the constants are not unique: any larger threshold, any smaller c_1 and any larger c_2 (or c in the definition of $\mathcal{O}$) also work.
- To prove that a statement is false, we proceed by contradiction: we suppose that the constants exist and we derive an inequality that cannot hold for arbitrarily large n.

10.1.2 Examples of asymptotic tight bounds

Example 10.1: asymptotic tight bound

- To prove that
$$-3\,n + \frac{1}{2}\,n^2 = \Theta(n^2)$$
we determine $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that
$$0 \leq c_1\,n^2 \leq -3\,n + \frac{1}{2}\,n^2 \leq c_2\,n^2, \qquad \forall n \geq n_1.$$
With $n \geq 1$, dividing by n^2 and since $c_1 > 0$, this is equivalent to
$$0 < c_1 \leq \frac{1}{2} - \frac{3}{n} \leq c_2, \qquad \forall n \geq n_1.$$
The middle term increases with n and it is always smaller than $\frac{1}{2}$. Thus, by choosing for example $c_1 = \frac{1}{14}$, $c_2 = \frac{1}{2}$ and $n_1 = 7$ (we selected $\frac{1}{14}$ since $\frac{1}{14} = \frac{1}{2} - \frac{3}{7}$), we have
$$0 \leq \frac{1}{14}\,n^2 \leq -3\,n + \frac{1}{2}\,n^2 \leq \frac{1}{2}\,n^2, \qquad \forall n \geq 7.$$
For $n = 6$ the lower bound fails, since $-18 + 18 = 0 < \frac{36}{14}$.



- The previous proof also proves that

$$-3\,n+\frac{1}{2}\,n^2=\mathcal{O}(n^2) \qquad \text{and} \qquad -3\,n+\frac{1}{2}\,n^2=\Omega(n^2).$$

Example 10.2: asymptotic tight bound

- To prove that

$$n^2\,(2-\sin n)=\Theta(n^2)$$

we determine $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

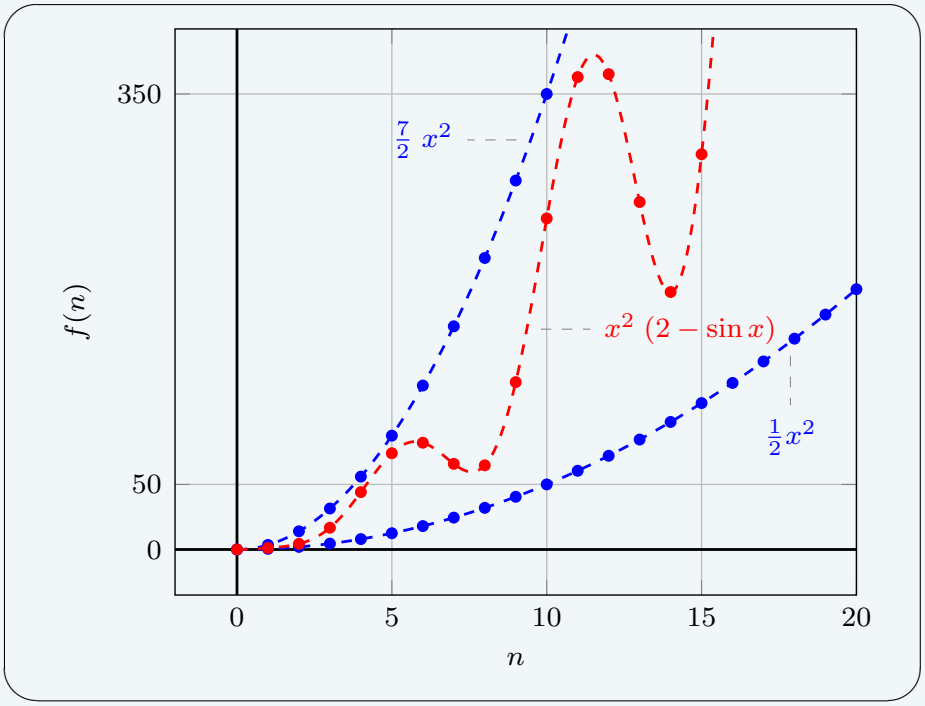
$$0 \leq c_1\,n^2 \leq n^2\,(2-\sin n) \leq c_2\,n^2, \qquad \forall n \geq n_1.$$

With $n \geq 1$, dividing by n^2 and since $c_1 > 0$, this is equivalent to

$$0 < c_1 \leq 2-\sin n \leq c_2, \qquad \forall n \geq n_1.$$

Since $-1 \leq \sin n \leq 1$, we have $1 \leq 2-\sin n \leq 3$. Thus, by choosing for example $c_1=\frac{1}{2}$, $c_2=\frac{7}{2}$ and $n_1=1$, we have

$$0 \leq \frac{1}{2}\,n^2 \leq n^2\,(2-\sin n) \leq \frac{7}{2}\,n^2, \qquad \forall n \geq 1.$$

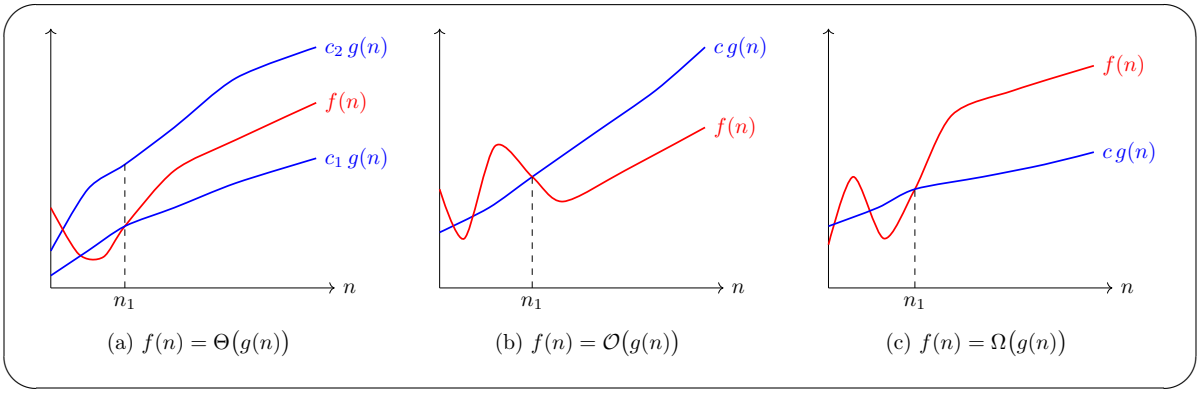


- The previous proof also proves that

$$n^2\,(2-\sin n) = \mathcal{O}(n^2) \qquad \text{and} \qquad n^2\,(2-\sin n) = \Omega(n^2).$$

10.1.3 Graphical interpretation

The next figure gives a graphical interpretation of the three notations.



(a) $f(n) = \Theta(g(n))$ (b) $f(n) = \mathcal{O}(g(n))$ (c) $f(n) = \Omega(g(n))$

- In (a), from n_1 on, the values of f lie between $c_1\,g(n)$ and $c_2\,g(n)$; in (b) they lie below $c\,g(n)$; in (c) they lie above $c\,g(n)$. Before n_1 nothing is required.
- In the figure, n_1 is the smallest possible threshold, but any larger value would also work. The figure also illustrates one possible choice of the constants c_1, c_2 and c, but other valid choices exist.

10.1.4 Θ versus $\mathcal{O}$ and Ω

Observation 10.5: Θ is the same as $\mathcal{O}$ and Ω

Given two functions f and g of a natural variable, we have

$$f(n) = \Theta\big(g(n)\big) \iff f(n) = \mathcal{O}\big(g(n)\big) \text{ and } f(n) = \Omega\big(g(n)\big). \tag{10.4}$$

In particular,

$$f(n) = \Theta\big(g(n)\big) \implies f(n) = \mathcal{O}\big(g(n)\big) \text{ and } f(n) = \Theta\big(g(n)\big) \implies f(n) = \Omega\big(g(n)\big). \tag{10.5}$$

Proof. If $f(n) = \Theta\big(g(n)\big)$, the two inequalities $0 \le f(n) \le c_2\, g(n)$ and $0 \le c_1\, g(n) \le f(n)$ contained in (10.1) give $f(n) = \mathcal{O}\big(g(n)\big)$ (with $c = c_2$) and $f(n) = \Omega\big(g(n)\big)$ (with $c = c_1$), with the same threshold n_1.

Conversely, let $0 \le f(n) \le c'\, g(n)$ for all $n \ge n_1'$, and $0 \le c''\, g(n) \le f(n)$ for all $n \ge n_1''$. By choosing $c_1 = c''$, $c_2 = c'$ and $n_1 = \max\{n_1', n_1''\}$, we obtain

$$0 \le c_1\, g(n) \le f(n) \le c_2\, g(n), \qquad \forall n \ge n_1,$$

that is, $f(n) = \Theta\big(g(n)\big)$. □

- It follows that "$f(n) = \Theta\big(g(n)\big)$" is a sufficient condition for "$f(n) = \mathcal{O}\big(g(n)\big)$", and "$f(n) = \mathcal{O}\big(g(n)\big)$" is a necessary condition for "$f(n) = \Theta\big(g(n)\big)$". By contraposition we also have
$$f(n) \ne \mathcal{O}\big(g(n)\big) \implies f(n) \ne \Theta\big(g(n)\big).$$
The implication does not work in the other direction (see the next two examples), that is,
$$f(n) = \mathcal{O}\big(g(n)\big) \not\Longrightarrow f(n) = \Theta\big(g(n)\big).$$

- In the same way, "$f(n) = \Theta\big(g(n)\big)$" is a sufficient condition for "$f(n) = \Omega\big(g(n)\big)$", and "$f(n) = \Omega\big(g(n)\big)$" is a necessary condition for "$f(n) = \Theta\big(g(n)\big)$". By contraposition we also have
$$f(n) \ne \Omega\big(g(n)\big) \implies f(n) \ne \Theta\big(g(n)\big),$$
and the implication does not work in the other direction (see the third example below):
$$f(n) = \Omega\big(g(n)\big) \not\Longrightarrow f(n) = \Theta\big(g(n)\big).$$

Example 10.3: asymptotic upper bound that is not tight

- To prove that
$$3\, n^2 = \mathcal{O}(n^3)$$

we determine $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq 3\,n^2 \leq c\,n^3, \qquad \forall n \geq n_1.$$

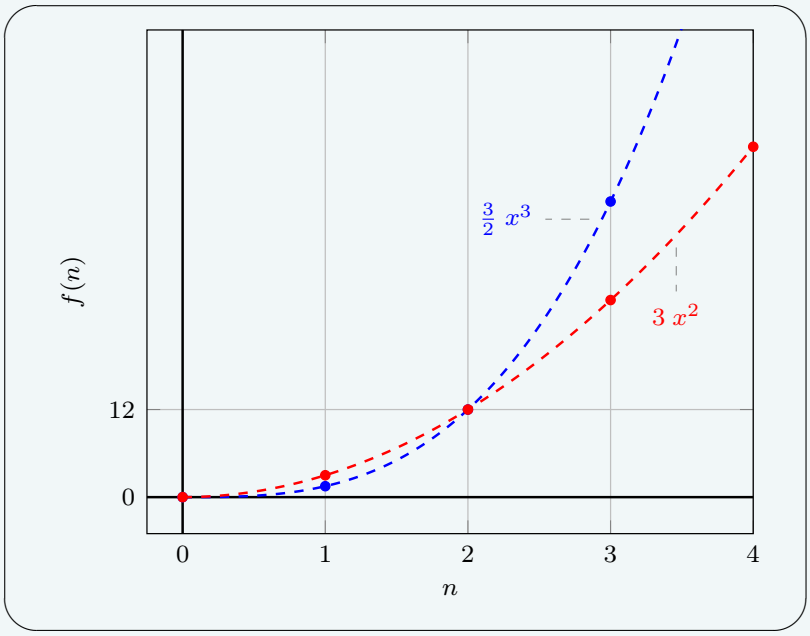
With $n \geq 1$, dividing by n^3, this is equivalent to

$$0 \leq \frac{3}{n} \leq c, \qquad \forall n \geq n_1.$$

Thus, by choosing for example $c = \frac{3}{2}$ and $n_1 = 2$, we have

$$0 \leq 3\,n^2 \leq \frac{3}{2}\,n^3, \qquad \forall n \geq 2.$$

(For $n = 1$ the inequality fails, since $3 > \frac{3}{2}$.)



- To prove that

$$3\,n^2 \neq \Theta(n^3)$$

we proceed by contradiction and we suppose that there exist $c_1 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq c_1\,n^3 \leq 3\,n^2, \qquad \forall n \geq n_1.$$

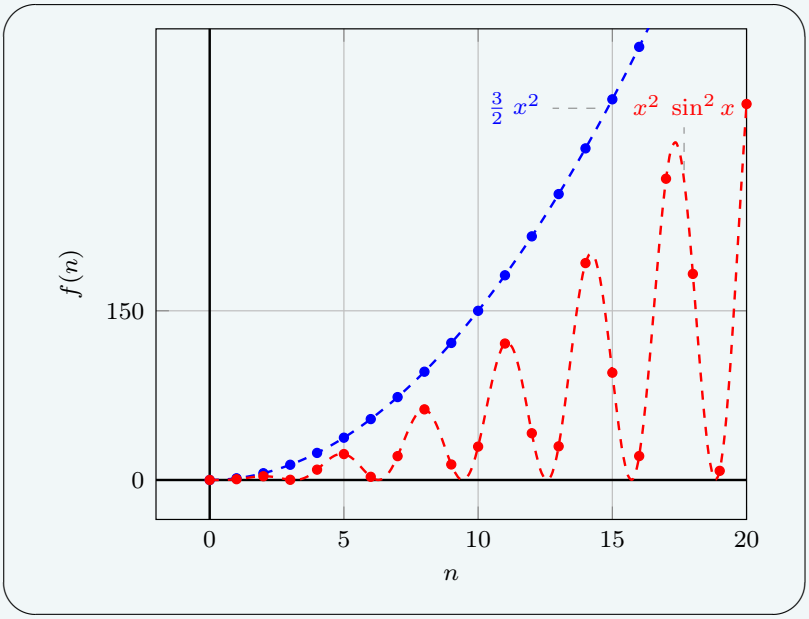
With $n \geq 1$, dividing by n^3 and since $c_1 > 0$, this is equivalent to

$$0 < c_1 \leq \frac{3}{n}, \qquad \forall n \geq n_1,$$

which cannot hold for arbitrarily large n, since c_1 is a positive constant (it fails for every $n > \frac{3}{c_1}$).

Example 10.4: asymptotic upper bound that is not tight

- To prove that
$$n^2 \ \sin^2 n = \mathcal{O}(n^2)$$
we determine $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that
$$0 \leq n^2 \ \sin^2 n \leq c\, n^2, \qquad \forall n \geq n_1.$$
With $n \geq 1$, dividing by n^2, this is equivalent to
$$0 \leq \sin^2 n \leq c, \qquad \forall n \geq n_1.$$
Since $0 \leq \sin^2 n \leq 1$, by choosing for example $c = \frac{3}{2}$ and $n_1 = 1$ we have
$$0 \leq n^2 \ \sin^2 n \leq \frac{3}{2}\, n^2, \qquad \forall n \geq 1.$$

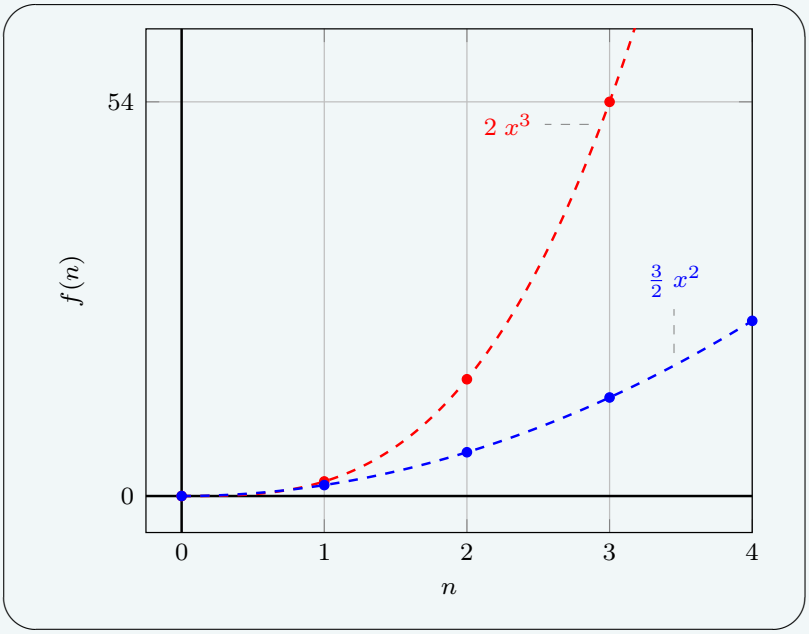


- To prove that
$$n^2 \ \sin^2 n \neq \Theta(n^2)$$
we proceed by contradiction and we suppose that there exist $c_1 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that
$$0 \leq c_1\, n^2 \leq n^2 \ \sin^2 n, \qquad \forall n \geq n_1.$$
With $n \geq 1$, dividing by n^2 and since $c_1 > 0$, this is equivalent to
$$0 < c_1 \leq \sin^2 n, \qquad \forall n \geq n_1,$$
which cannot hold for arbitrarily large n, since c_1 is a positive constant.[a]

[a] Since π is irrational, the values $n \bmod \pi$, $n \in \mathbb{N}$, are dense in $[0, \pi]$. Therefore, for every $\varepsilon \in \mathbb{R}_{>0}$ there exist arbitrarily large integers n such that $|\sin n| < \varepsilon$. By choosing $\varepsilon = \frac{\sqrt{c_1}}{2}$, we obtain $\sin^2 n < \frac{c_1}{4} < c_1$, which contradicts $c_1 \leq \sin^2 n$ for all sufficiently large n.

Example 10.5: asymptotic lower bound that is not tight

- To prove that
$$2\,n^3 = \Omega(n^2)$$
we determine $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that
$$0 \leq c\,n^2 \leq 2\,n^3, \qquad \forall n \geq n_1.$$
With $n \geq 1$, dividing by n^2 and since $c > 0$, this is equivalent to
$$0 < c \leq 2\,n, \qquad \forall n \geq n_1.$$
Thus, by choosing for example $c = \frac{3}{2}$ and $n_1 = 1$, we have
$$0 \leq \frac{3}{2}\,n^2 \leq 2\,n^3, \qquad \forall n \geq 1.$$



- To prove that
$$2\,n^3 \neq \Theta(n^2)$$
we proceed by contradiction and we suppose that there exist $c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that
$$0 \leq 2\,n^3 \leq c_2\,n^2, \qquad \forall n \geq n_1.$$
With $n \geq 1$, dividing by n^2, this is equivalent to
$$0 \leq 2\,n \leq c_2, \qquad \forall n \geq n_1,$$
which cannot hold for arbitrarily large n, since c_2 is a constant (it fails for every $n > \frac{c_2}{2}$).

10.1.5 Sum and maximum of two functions

Observation 10.6: adding the bound to the function

Given two functions f and g of a natural variable, we have

$$f(n) = \Theta\big(g(n)\big) \implies f(n) + g(n) = \Theta\big(g(n)\big). \tag{10.6}$$

Proof. If $f(n) = \Theta\big(g(n)\big)$, there exist $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq c_1\, g(n) \leq f(n) \leq c_2\, g(n), \qquad \forall n \geq n_1.$$

Moreover, since $c_1 > 0$, we have $g(n) \geq 0$ for all $n \geq n_1$. Then, by adding $g(n)$, we get

$$0 \leq \underbrace{c_1\, g(n) + g(n)}_{=\underbrace{(c_1+1)}_{>0}\, g(n)} \leq f(n) + g(n) \leq \underbrace{c_2\, g(n) + g(n)}_{=\underbrace{(c_2+1)}_{>0}\, g(n)}, \qquad \forall n \geq n_1.$$

Accordingly, $f(n) + g(n) = \Theta\big(g(n)\big)$, with the constants $c_1 + 1$ and $c_2 + 1$ and the same threshold n_1. □

Example 10.6: asymptotic tight bound of a sum

- In Example 10.1 we proved that

  $$-3\,n + \frac{1}{2}\,n^2 = \Theta(n^2)$$

  with $c_1 = \frac{1}{14}$, $c_2 = \frac{1}{2}$ and $n_1 = 7$. Then, according to (10.6), we have

  $$-3\,n + \frac{1}{2}\,n^2 + n^2 = \Theta(n^2),$$

  with the constants $c_1 + 1 = \frac{15}{14}$, $c_2 + 1 = \frac{3}{2}$ and $n_1 = 7$.

- In Example 10.2 we proved that

  $$n^2\,(2 - \sin n) = \Theta(n^2)$$

  with $c_1 = \frac{1}{2}$, $c_2 = \frac{7}{2}$ and $n_1 = 1$. Then, according to (10.6), we have

  $$n^2\,(2 - \sin n) + n^2 = \Theta(n^2),$$

  with the constants $\frac{3}{2}$, $\frac{9}{2}$ and $n_1 = 1$.

- By the implication (10.6), the statement "$f(n) = \Theta\big(g(n)\big)$" is a sufficient condition for "$f(n) + g(n) = \Theta\big(g(n)\big)$", and "$f(n) + g(n) = \Theta\big(g(n)\big)$" is a necessary condition for "$f(n) = \Theta\big(g(n)\big)$". By contraposition of (10.6) we also have

  $$f(n) + g(n) \neq \Theta\big(g(n)\big) \implies f(n) \neq \Theta\big(g(n)\big).$$

The implication (10.6) does not work in the other direction (see the next counterexample), that is,

$$f(n) + g(n) = \Theta\big(g(n)\big) \quad \not\Rightarrow \quad f(n) = \Theta\big(g(n)\big).$$

Example 10.7: the converse of Observation 10.6 is false

Let $f(n) = 10\,n$ and $g(n) = \frac{1}{10}\,n^2$.

- To prove that

$$f(n) + g(n) = 10\,n + \frac{1}{10}\,n^2 = \Theta\Big(\frac{1}{10}\,n^2\Big) = \Theta\big(g(n)\big)$$

we determine $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

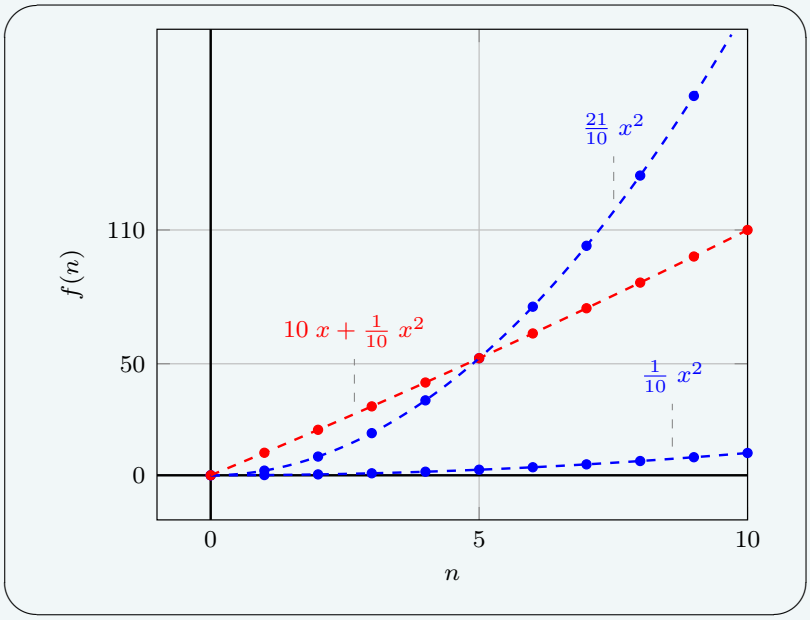
$$0 \le c_1\,\frac{1}{10}\,n^2 \le 10\,n + \frac{1}{10}\,n^2 \le c_2\,\frac{1}{10}\,n^2, \qquad \forall n \ge n_1.$$

With $n \ge 1$, dividing by $\frac{1}{10}\,n^2$ and since $c_1 > 0$, this is equivalent to

$$0 < c_1 \le \frac{100}{n} + 1 \le c_2, \qquad \forall n \ge n_1.$$

The middle term decreases with n and it is always larger than 1; for $n = 5$ it is equal to 21. Thus, by choosing for example $c_1 = 1$, $c_2 = 21$ and $n_1 = 5$, we have

$$0 \le \frac{1}{10}\,n^2 \le 10\,n + \frac{1}{10}\,n^2 \le \frac{21}{10}\,n^2, \qquad \forall n \ge 5.$$



- To prove that

$$f(n) = 10\,n \neq \Theta\Big(\frac{1}{10}\,n^2\Big) = \Theta\big(g(n)\big)$$

we proceed by contradiction and we suppose that there exist $c_1 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \le c_1\,\frac{1}{10}\,n^2 \le 10\,n, \qquad \forall n \ge n_1.$$

With $n \geq 1$, dividing by $\frac{1}{10}\, n^2$ and since $c_1 > 0$, this is equivalent to

$$0 < c_1 \leq \frac{100}{n}, \qquad \forall n \geq n_1,$$

which cannot hold for arbitrarily large n, since c_1 is a positive constant.

When we add two functions, the asymptotic behaviour of the sum is determined by the larger of the two. The next two observations make this precise.

Observation 10.7: sum and maximum (1)

Given two asymptotically non-negative functions f and g of a natural variable, we have

$$f(n) + g(n) = \Theta\big(\max\{f(n), g(n)\}\big).$$

Proof. Since f and g are asymptotically non-negative, there exists $\tilde{n} \in \mathbb{N}$ such that $f(n) \geq 0$ and $g(n) \geq 0$ for all $n \geq \tilde{n}$. Then the sum is at least the larger of the two terms and at most twice the larger of the two terms:

$$0 \leq \max\{f(n), g(n)\} \leq f(n) + g(n) \leq 2\ \max\{f(n), g(n)\}, \qquad \forall n \geq \tilde{n}.$$

This shows that the constants $c_1 = 1$, $c_2 = 2$ and $n_1 = \tilde{n}$ satisfy

$$0 \leq c_1\ \max\{f(n), g(n)\} \leq f(n) + g(n) \leq c_2\ \max\{f(n), g(n)\}, \qquad \forall n \geq n_1.$$

Therefore, $f(n) + g(n) = \Theta\big(\max\{f(n), g(n)\}\big)$. □

Observation 10.8: sum and maximum (2)

Given two asymptotically non-negative functions f and g of a natural variable, we have

$$\max\{f(n), g(n)\} = \Theta\big(f(n) + g(n)\big).$$

Proof. Let $\tilde{n} \in \mathbb{N}$ be such that $f(n) \geq 0$ and $g(n) \geq 0$ for all $n \geq \tilde{n}$. The larger of two non-negative numbers is at least their average and at most their sum:

$$0 \leq \frac{1}{2}\,\big(f(n) + g(n)\big) \leq \max\{f(n), g(n)\} \leq f(n) + g(n), \qquad \forall n \geq \tilde{n}.$$

This shows that the constants $c_1 = \frac{1}{2}$, $c_2 = 1$ and $n_1 = \tilde{n}$ satisfy

$$0 \leq c_1\,\big(f(n) + g(n)\big) \leq \max\{f(n), g(n)\} \leq c_2\,\big(f(n) + g(n)\big), \qquad \forall n \geq n_1.$$

Therefore, $\max\{f(n), g(n)\} = \Theta\big(f(n) + g(n)\big)$. □

Example 10.8: sum and maximum

By Observations 10.7 and 10.8 we have

$$\frac{1}{2}\,n^2 + 2^n = \Theta\left(\max\left\{\frac{1}{2}\,n^2, 2^n\right\}\right) \qquad \text{and} \qquad \max\left\{\frac{1}{2}\,n^2, 2^n\right\} = \Theta\left(\frac{1}{2}\,n^2 + 2^n\right).$$

In this case $\max\left\{\frac{1}{2}\,n^2, 2^n\right\} = 2^n$ for every $n \in \mathbb{N}$, so that $\frac{1}{2}\,n^2 + 2^n = \Theta(2^n)$, with $c_1 = 1$, $c_2 = 2$ and $n_1 = 0$.

10.2 o and ω notations

The asymptotic bounds expressed by the $\mathcal{O}$- and Ω-notations may or may not be asymptotically tight. We use the o-notation to denote an asymptotic upper bound that is **not asymptotically tight**, and the ω-notation to denote an asymptotic lower bound that is **not asymptotically tight**.

Example 10.9: asymptotically tight and non-tight upper bounds

- We have $3n^3 = \mathcal{O}(n^3)$ and $3n^3 = \Theta(n^3)$: n^3 is an asymptotic upper bound for $3n^3$, and it is also an asymptotically tight bound.
- We have $3n^3 = \mathcal{O}(n^4)$ but $3n^3 \neq \Theta(n^4)$: n^4 is an asymptotic upper bound for $3n^3$, but it is not an asymptotically tight bound.

Definition 10.9: o-notation

Given two functions f and g of a natural variable, we write $f(n) = o\big(g(n)\big)$ if for all $c \in \mathbb{R}_{>0}$ there exists $n_1 \in \mathbb{N}$ such that

$$0 \leq f(n) < c\,g(n), \qquad \forall n \geq n_1. \tag{10.7}$$

- If $f(n) = o\big(g(n)\big)$, then, for sufficiently large values of n, the function f is bounded from above by the function g scaled by **any** positive constant, that is, the function f becomes insignificant relative to the function g.
- In this case we say that the function g is an **asymptotic upper bound that is not tight** for the function f.
- The difference with the $\mathcal{O}$-notation is in the quantifier on the constant: for $\mathcal{O}$ the inequality must hold for **one** constant c, for o it must hold for **every** constant c. The threshold n_1 depends on c: the smaller c, the larger, in general, n_1.

Definition 10.10: ω-notation

Given two functions f and g of a natural variable, we write $f(n) = \omega\big(g(n)\big)$ if for all

$c \in \mathbb{R}_{>0}$ there exists $n_1 \in \mathbb{N}$ such that

$$0 \leq c\, g(n) < f(n), \qquad \forall n \geq n_1. \tag{10.8}$$

- If $f(n) = \omega\big(g(n)\big)$, then, for sufficiently large values of n, the function f is bounded from below by the function g scaled by **any** positive constant, that is, the function f becomes arbitrarily large relative to the function g.
- In this case we say that the function g is an **asymptotic lower bound that is not tight** for the function f.

If $f(n) = o\big(g(n)\big)$ or $f(n) = \omega\big(g(n)\big)$ then, since $c > 0$, we have $g(n) \geq 0$ and $f(n) \geq 0$ for all $n \geq n_1$. Therefore, the two functions f and g are **asymptotically non-negative**.

Example 10.10: asymptotic upper bound that is not tight

- To prove that

  $$3n = o(n^2)$$

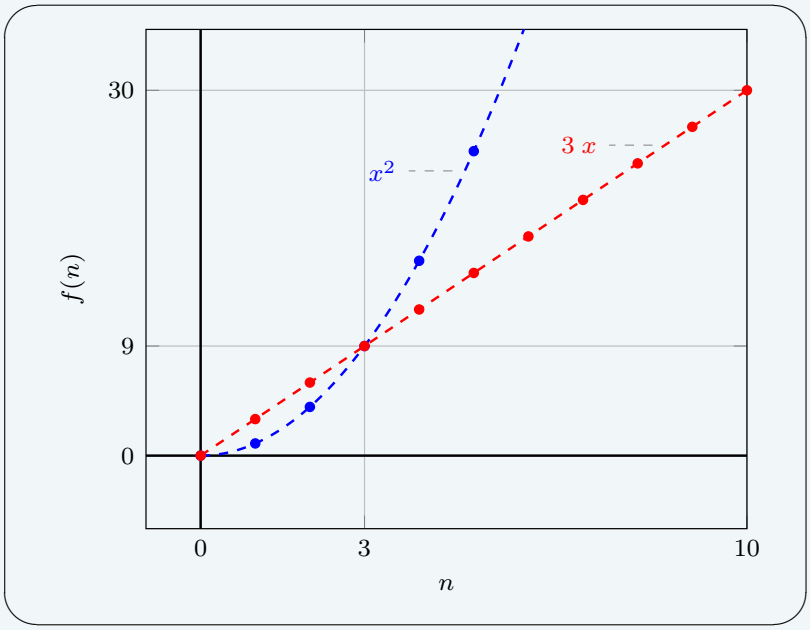
  we need to verify that for every $c \in \mathbb{R}_{>0}$ there exists $n_1 \in \mathbb{N}$ such that

  $$0 \leq 3n < c\, n^2, \qquad \forall n \geq n_1.$$

  For $n \geq 1$, dividing by n^2, the condition is equivalent to

  $$0 \leq \frac{3}{n} < c, \qquad \forall n \geq n_1.$$

  This holds whenever $n > \frac{3}{c}$, so we can choose any $n_1 > \frac{3}{c}$, for example $n_1 = \lfloor \frac{3}{c} \rfloor + 1$. For instance, $c = 1$ gives $n_1 = 4$, $c = \frac{1}{2}$ gives $n_1 = 7$, and $c = \frac{1}{10}$ gives $n_1 = 31$.



- The previous proof also proves that

$$3n = \mathcal{O}(n^2)$$

(it is enough to take one value of c, for example $c = 1$ with $n_1 = 4$).

- To prove that

$$3\,n \neq \Theta(n^2)$$

we proceed by contradiction and we suppose that there exist $c_1 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq c_1\, n^2 \leq 3\, n, \qquad \forall n \geq n_1.$$

With $n \geq 1$, dividing by n^2 and since $c_1 > 0$, this is equivalent to

$$0 < c_1 \leq \frac{3}{n}, \qquad \forall n \geq n_1,$$

which cannot hold for arbitrarily large n, since c_1 is a positive constant.

Example 10.11: asymptotic lower bound that is not tight

- To prove that

$$\frac{1}{5}\, n^2 = \omega(n)$$

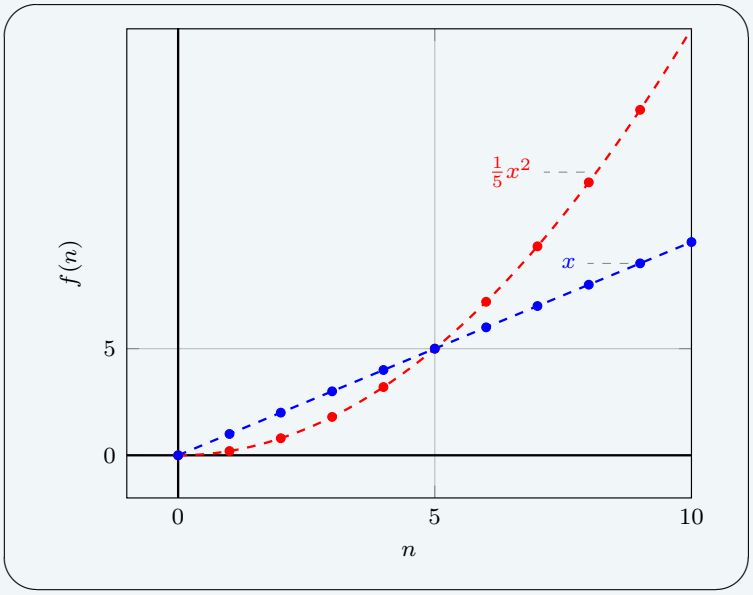
we need to verify that for every $c \in \mathbb{R}_{>0}$ there exists $n_1 \in \mathbb{N}$ such that

$$0 \leq c\, n < \frac{1}{5}\, n^2, \qquad \forall n \geq n_1.$$

For $n \geq 1$, dividing by n, the condition is equivalent to

$$0 < c < \frac{1}{5}\, n, \qquad \forall n \geq n_1.$$

This holds whenever $n > 5c$, so we can choose any $n_1 > 5c$, for example $n_1 = \lfloor 5c \rfloor + 1$. For instance, $c = 1$ gives $n_1 = 6$ and $c = 10$ gives $n_1 = 51$.



- The previous proof also proves that

$$\frac{1}{5}n^2 = \Omega(n).$$

- To prove that

$$\frac{1}{5}n^2 \neq \Theta(n)$$

we proceed by contradiction and we suppose that there exist $c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq \frac{1}{5}n^2 \leq c_2\, n, \qquad \forall n \geq n_1.$$

With $n \geq 1$, dividing by n, this is equivalent to

$$0 \leq \frac{n}{5} \leq c_2, \qquad \forall n \geq n_1,$$

which cannot hold for arbitrarily large n, since c_2 is a constant.

10.3 Limits and asymptotic notations

When the ratio $\frac{f(n)}{g(n)}$ has a limit, its value tells us immediately which notations hold. In this section the functions f and g are **asymptotically positive**, so that the ratio is defined for all sufficiently large n. The limits are computed with the rules of Mathematical Analysis 1.

10.3.1 Limits and the notations o and ω

Observation 10.11: limit equal to zero

Given two asymptotically positive functions f and g of a natural variable, we have

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = 0^+ \iff f(n) = o\big(g(n)\big). \tag{10.9}$$

Proof. Let $\tilde{n} \in \mathbb{N}$ be such that $f(n) > 0$ and $g(n) > 0$ for all $n \geq \tilde{n}$.

- If $\lim_{n\to+\infty} \frac{f(n)}{g(n)} = 0^+$, then for all $\varepsilon \in \mathbb{R}_{>0}$ there exists $n(\varepsilon) \in \mathbb{N}$, which we can take with $n(\varepsilon) \geq \tilde{n}$, such that

  $$0 \leq \frac{f(n)}{g(n)} < \varepsilon, \quad \forall n \geq n(\varepsilon) \qquad \Longrightarrow \qquad 0 \leq f(n) < \varepsilon\, g(n), \quad \forall n \geq n(\varepsilon),$$

  where we multiplied by $g(n) > 0$. By choosing $c = \varepsilon$ and $n_1 = n(\varepsilon)$, this is the definition (10.7), so $f(n) = o\big(g(n)\big)$.

- If $f(n) = o\big(g(n)\big)$, then for all $c \in \mathbb{R}_{>0}$ there exists $n_1 \in \mathbb{N}$, which we can take with $n_1 \geq \tilde{n}$, such that

  $$0 \leq f(n) < c\, g(n), \quad \forall n \geq n_1 \qquad \Longrightarrow \qquad 0 \leq \frac{f(n)}{g(n)} < c, \quad \forall n \geq n_1,$$

  where we divided by $g(n) > 0$. Since c is arbitrary, this is the definition of $\lim_{n\to+\infty} \frac{f(n)}{g(n)} = 0^+$.

□

- If the functions f and g are asymptotically positive, "$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = 0^+$" is a necessary and sufficient condition for "$f(n) = o\big(g(n)\big)$".

Observation 10.12: o implies $\mathcal{O}$

Given two functions f and g of a natural variable, we have

$$f(n) = o\big(g(n)\big) \implies f(n) = \mathcal{O}\big(g(n)\big). \tag{10.10}$$

Proof. The definition (10.7) holds for every $c \in \mathbb{R}_{>0}$; in particular it holds for $c = 1$: there exists $n_1 \in \mathbb{N}$ such that $0 \leq f(n) < g(n) \leq 1 \cdot g(n)$ for all $n \geq n_1$. This is the definition (10.2) with $c = 1$. □

- It follows that "$f(n) = o\big(g(n)\big)$" is a sufficient condition for "$f(n) = \mathcal{O}\big(g(n)\big)$", and "$f(n) = \mathcal{O}\big(g(n)\big)$" is a necessary condition for "$f(n) = o\big(g(n)\big)$". By contraposition of (10.10) we also have

$$f(n) \neq \mathcal{O}\big(g(n)\big) \implies f(n) \neq o\big(g(n)\big).$$

The implication does not work in the other direction (see the next counterexample), that is,

$$f(n) = \mathcal{O}\big(g(n)\big) \not\Rightarrow f(n) = o\big(g(n)\big).$$

Example 10.12: an upper bound that is not an o

- To prove that

$$3\,n^3 = \mathcal{O}(n^3)$$

we determine $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq 3\,n^3 \leq c\,n^3, \qquad \forall n \geq n_1.$$

With $n \geq 1$, dividing by n^3, this is equivalent to

$$0 \leq 3 \leq c, \qquad \forall n \geq n_1.$$

Thus, by choosing for example $c = \frac{7}{2}$ and $n_1 = 1$, we have

$$0 \leq 3\,n^3 \leq \frac{7}{2}\,n^3, \qquad \forall n \geq 1.$$

- To prove that

$$3\,n^3 \neq o(n^3)$$

we compute the limit

$$\lim_{n \to +\infty} \frac{3\,n^3}{n^3} = 3 \neq 0^+$$

and we apply (10.9).

Observation 10.13: limit equal to $+\infty$

Given two asymptotically positive functions f and g of a natural variable, we have

$$\lim_{n \to +\infty} \frac{f(n)}{g(n)} = +\infty \iff f(n) = \omega\big(g(n)\big). \tag{10.11}$$

Proof. Let $\tilde{n} \in \mathbb{N}$ be such that $f(n) > 0$ and $g(n) > 0$ for all $n \geq \tilde{n}$.

- If $\lim_{n\to+\infty} \dfrac{f(n)}{g(n)} = +\infty$, then for all $\lambda \in \mathbb{R}_{>0}$ there exists $n(\lambda) \in \mathbb{N}$, which we can take with $n(\lambda) \geq \tilde{n}$, such that

$$\frac{f(n)}{g(n)} > \lambda, \quad \forall n \geq n(\lambda) \qquad \Longrightarrow \qquad 0 \leq \lambda\, g(n) < f(n), \quad \forall n \geq n(\lambda).$$

  By choosing $c = \lambda$ and $n_1 = n(\lambda)$, this is the definition (10.8), so $f(n) = \omega\big(g(n)\big)$.

- If $f(n) = \omega\big(g(n)\big)$, then for all $c \in \mathbb{R}_{>0}$ there exists $n_1 \in \mathbb{N}$, which we can take with $n_1 \geq \tilde{n}$, such that

$$0 \leq c\, g(n) < f(n), \quad \forall n \geq n_1 \qquad \Longrightarrow \qquad \frac{f(n)}{g(n)} > c, \quad \forall n \geq n_1.$$

  Since c is arbitrary, this is the definition of $\lim_{n\to+\infty} \dfrac{f(n)}{g(n)} = +\infty$.

□

- If the functions f and g are asymptotically positive, "$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = +\infty$" is a necessary and sufficient condition for "$f(n) = \omega\big(g(n)\big)$".

Observation 10.14: ω implies Ω

Given two functions f and g of a natural variable, we have

$$f(n) = \omega\big(g(n)\big) \quad \Longrightarrow \quad f(n) = \Omega\big(g(n)\big). \tag{10.12}$$

Proof. The definition (10.8) holds for every $c \in \mathbb{R}_{>0}$; in particular it holds for $c = 1$: there exists $n_1 \in \mathbb{N}$ such that $0 \leq 1 \cdot g(n) < f(n)$ for all $n \geq n_1$. This is the definition (10.3) with $c = 1$. □

- It follows that "$f(n) = \omega\big(g(n)\big)$" is a sufficient condition for "$f(n) = \Omega\big(g(n)\big)$", and "$f(n) = \Omega\big(g(n)\big)$" is a necessary condition for "$f(n) = \omega\big(g(n)\big)$". By contraposition of (10.12) we also have

$$f(n) \neq \Omega\big(g(n)\big) \quad \Longrightarrow \quad f(n) \neq \omega\big(g(n)\big).$$

  The implication does not work in the other direction (see the next counterexample), that is,

$$f(n) = \Omega\big(g(n)\big) \quad \not\Rightarrow \quad f(n) = \omega\big(g(n)\big).$$

Example 10.13: a lower bound that is not an ω

- To prove that
$$3\,n^3 = \Omega(n^3)$$
we determine $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that
$$0 \leq c\,n^3 \leq 3\,n^3, \qquad \forall n \geq n_1.$$
With $n \geq 1$, dividing by n^3 and since $c > 0$, this is equivalent to
$$0 < c \leq 3, \qquad \forall n \geq n_1.$$
Thus, by choosing for example $c = \frac{5}{2}$ and $n_1 = 1$, we have
$$0 \leq \frac{5}{2}\,n^3 \leq 3\,n^3, \qquad \forall n \geq 1.$$

- To prove that
$$3\,n^3 \neq \omega(n^3)$$
we compute the limit
$$\lim_{n\to+\infty} \frac{3\,n^3}{n^3} = 3 \neq +\infty$$
and we apply (10.11).

10.3.2 Limits and the notations Θ, $\mathcal{O}$ and Ω

Observation 10.15: finite positive limit

Given two asymptotically positive functions f and g of a natural variable, we have
$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \implies f(n) = \Theta\big(g(n)\big). \tag{10.13}$$

Proof. If $\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0}$, then for all $\varepsilon \in \mathbb{R}_{>0}$ there exists $n(\varepsilon) \in \mathbb{N}$ such that $g(n) > 0$ and
$$\ell - \varepsilon < \frac{f(n)}{g(n)} < \ell + \varepsilon, \qquad \forall n \geq n(\varepsilon).$$
Since $\ell > 0$, we can choose ε small enough to have $\ell - \varepsilon > 0$, for example $\varepsilon = \frac{\ell}{2}$. By setting $c_1 = \ell - \varepsilon = \frac{\ell}{2} > 0$ and $c_2 = \ell + \varepsilon = \frac{3\ell}{2} > 0$, and multiplying by $g(n) > 0$, we have
$$0 < c_1\,g(n) < f(n) < c_2\,g(n), \qquad \forall n \geq n(\varepsilon),$$
and, as a consequence, $f(n) = \Theta\big(g(n)\big)$ with $n_1 = n(\varepsilon)$. □

- It follows that "$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0}$" is a sufficient condition for "$f(n) = \Theta\big(g(n)\big)$", and "$f(n) = \Theta\big(g(n)\big)$" is a necessary condition for "$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0}$". By

contraposition of (10.13) we also have

$$f(n) \neq \Theta\big(g(n)\big) \implies \text{the limit of } \frac{f(n)}{g(n)} \text{ is not a number } \ell \in \mathbb{R}_{>0}.$$

The implication does not work in the other direction. A counterexample is given by Example 10.2:

$$n^2\,(2-\sin n) = \Theta(n^2), \quad \text{but} \quad \lim_{n\to+\infty} \frac{n^2\,(2-\sin n)}{n^2} = \lim_{n\to+\infty} (2-\sin n) \text{ does not exist.}$$

Accordingly, we have

$$f(n) = \Theta\big(g(n)\big) \not\implies \lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0}.$$

It is possible that $f(n) = \Theta\big(g(n)\big)$ even though $\frac{f(n)}{g(n)}$ does not converge to a limit. In this case the limit criterion cannot be applied, and we use the definition.

Example 10.14: asymptotic tight bound without a limit

To prove that

$$(2+(-1)^n)\,n^2 = \Theta(n^2)$$

we determine $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

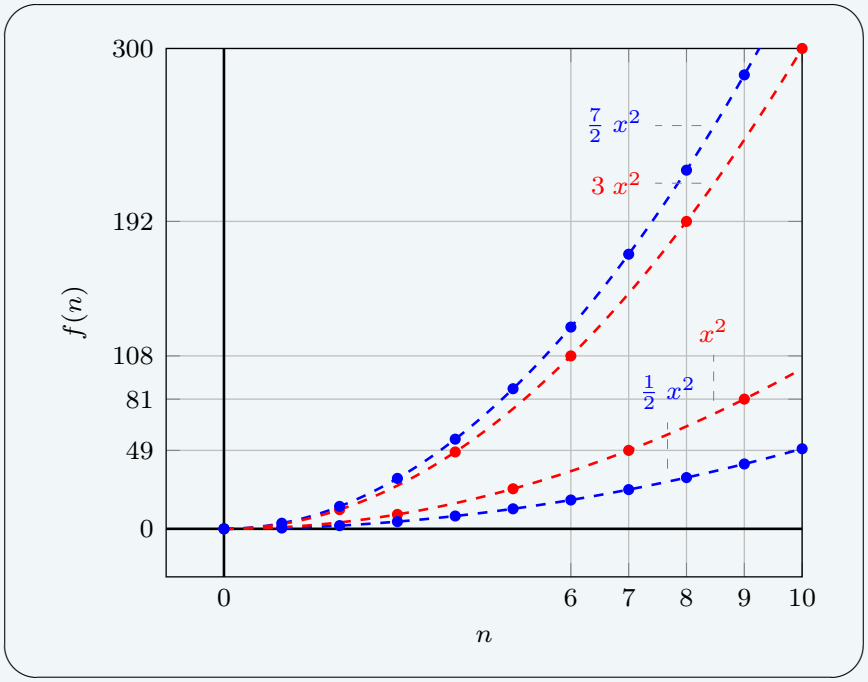
$$0 \leq c_1\,n^2 \leq (2+(-1)^n)\,n^2 \leq c_2\,n^2, \qquad \forall n \geq n_1.$$

Since

$$2+(-1)^n = \begin{cases} 1 & \text{if } n \text{ is odd,} \\ 3 & \text{if } n \text{ is even,} \end{cases} \qquad \text{we have} \qquad (2+(-1)^n)\,n^2 = \begin{cases} n^2 & \text{if } n \text{ is odd,} \\ 3\,n^2 & \text{if } n \text{ is even.} \end{cases}$$

Thus, by choosing for example $c_1 = \frac{1}{2}$, $c_2 = \frac{7}{2}$ and $n_1 = 1$, we have

$$0 \leq \frac{1}{2}\,n^2 \leq (2+(-1)^n)\,n^2 \leq \frac{7}{2}\,n^2, \qquad \forall n \geq 1.$$



On the other hand,

$$\lim_{n\to+\infty} \frac{(2+(-1)^n)\,n^2}{n^2} = \lim_{n\to+\infty} (2+(-1)^n) \text{ does not exist,}$$

since the ratio is 1 for all odd n and 3 for all even n.

Observation 10.16: limit criterion for $\mathcal{O}$

Given two asymptotically positive functions f and g of a natural variable, we have

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \text{ or } 0^+ \implies f(n) = \mathcal{O}\big(g(n)\big). \tag{10.14}$$

Proof. By Observations 10.15, 10.5, 10.11 and 10.12, we have

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \implies f(n) = \Theta\big(g(n)\big) \implies f(n) = \mathcal{O}\big(g(n)\big),$$

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = 0^+ \iff f(n) = o\big(g(n)\big) \implies f(n) = \mathcal{O}\big(g(n)\big).$$

In both cases $f(n) = \mathcal{O}\big(g(n)\big)$. □

- It follows that "$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0}$ or 0^+" is a sufficient condition for "$f(n) = \mathcal{O}\big(g(n)\big)$". By contraposition of (10.14) we also have

$$f(n) \neq \mathcal{O}\big(g(n)\big) \implies \text{the limit of } \frac{f(n)}{g(n)} \text{ is neither a number } \ell \in \mathbb{R}_{>0} \text{ nor } 0^+.$$

The implication does not work in the other direction. A counterexample is given by Example 10.4:

$$n^2 \sin^2 n = \mathcal{O}(n^2), \qquad \text{but} \qquad \lim_{n\to+\infty} \frac{n^2 \sin^2 n}{n^2} = \lim_{n\to+\infty} \sin^2 n \text{ does not exist.}$$

Accordingly, we have

$$f(n) = \mathcal{O}\big(g(n)\big) \quad \not\Longrightarrow \quad \lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \text{ or } 0^+.$$

It is possible that $f(n) = \mathcal{O}\big(g(n)\big)$ even though $\frac{f(n)}{g(n)}$ does not converge to a limit.

Observation 10.17: limit criterion for Ω

Given two asymptotically positive functions f and g of a natural variable, we have

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \text{ or } +\infty \implies f(n) = \Omega\big(g(n)\big). \tag{10.15}$$

Proof. By Observations 10.15, 10.5, 10.13 and 10.14, we have

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \implies f(n) = \Theta\big(g(n)\big) \implies f(n) = \Omega\big(g(n)\big),$$

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = +\infty \iff f(n) = \omega\big(g(n)\big) \implies f(n) = \Omega\big(g(n)\big).$$

In both cases $f(n) = \Omega\big(g(n)\big)$. □

- It follows that "$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0}$ or $+\infty$" is a sufficient condition for "$f(n) = \Omega\big(g(n)\big)$". By contraposition of (10.15) we also have

$$f(n) \neq \Omega\big(g(n)\big) \implies \text{the limit of } \frac{f(n)}{g(n)} \text{ is neither a number } \ell \in \mathbb{R}_{>0} \text{ nor } +\infty.$$

The implication does not work in the other direction. A counterexample is given by Example 10.14: since $(2 + (-1)^n)\, n^2 = \Theta(n^2)$, by Observation 10.5 we have

$$(2+(-1)^n)n^2 = \Omega(n^2), \quad \text{but} \quad \lim_{n\to+\infty} \frac{(2 + (-1)^n)\, n^2}{n^2} = \lim_{n\to+\infty} (2+(-1)^n) \text{ does not exist.}$$

Accordingly, we have

$$f(n) = \Omega\big(g(n)\big) \quad \not\Longrightarrow \quad \lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \text{ or } +\infty.$$

It is possible that $f(n) = \Omega\big(g(n)\big)$ even though $\frac{f(n)}{g(n)}$ does not converge to a limit.

Example 10.15: asymptotic bounds from limits

- We have
$$\lim_{n\to+\infty} \frac{6\,n - n^3 + 2\,n^4}{n^4} = \lim_{n\to+\infty} \left(\underbrace{\frac{6}{n^3}}_{\to 0} - \underbrace{\frac{1}{n}}_{\to 0} + 2 \right) = 2 \in \mathbb{R}_{>0}.$$
Then, according to (10.14), we have $6\,n - n^3 + 2\,n^4 = \mathcal{O}(n^4)$ and, according to (10.15), we have $6\,n - n^3 + 2\,n^4 = \Omega(n^4)$. Finally, according to (10.13), we also have $6\,n - n^3 + 2\,n^4 = \Theta(n^4)$.

- We have
$$\lim_{n\to+\infty} \frac{n + 3\,n^3}{n^4} = \lim_{n\to+\infty} \left(\underbrace{\frac{1}{n^3}}_{\to 0} + \underbrace{\frac{3}{n}}_{\to 0} \right) = 0^+.$$
Then, according to (10.9), we have $n + 3\,n^3 = o(n^4)$. Moreover, according to (10.10) or to (10.14), we have $n + 3\,n^3 = \mathcal{O}(n^4)$.

- We have
$$\lim_{n\to+\infty} \frac{n + 3\,n^4}{n^3} = \lim_{n\to+\infty} \left(\underbrace{\frac{1}{n^2}}_{\to 0} + 3\,n \right) = +\infty.$$
Then, according to (10.11), we have $n + 3\,n^4 = \omega(n^3)$. Moreover, according to (10.12) or to (10.15), we have $n + 3\,n^4 = \Omega(n^3)$.

10.4 Relational properties of asymptotic notations

Proposition 10.18: relational properties

Given three functions f, g and h of a natural variable, the following properties hold.

1. **Transitivity**:
$$\begin{aligned}
&f(n) = o\big(g(n)\big) \text{ and } g(n) = o\big(h(n)\big) \implies f(n) = o\big(h(n)\big),\\
&f(n) = \mathcal{O}\big(g(n)\big) \text{ and } g(n) = \mathcal{O}\big(h(n)\big) \implies f(n) = \mathcal{O}\big(h(n)\big),\\
&f(n) = \Theta\big(g(n)\big) \text{ and } g(n) = \Theta\big(h(n)\big) \implies f(n) = \Theta\big(h(n)\big),\\
&f(n) = \Omega\big(g(n)\big) \text{ and } g(n) = \Omega\big(h(n)\big) \implies f(n) = \Omega\big(h(n)\big),\\
&f(n) = \omega\big(g(n)\big) \text{ and } g(n) = \omega\big(h(n)\big) \implies f(n) = \omega\big(h(n)\big).
\end{aligned}$$

2. **Reflexivity** (for an asymptotically non-negative function f):
$$f(n) = \Theta\big(f(n)\big), \qquad f(n) = \mathcal{O}\big(f(n)\big), \qquad f(n) = \Omega\big(f(n)\big).$$

3. **Symmetry**:
$$f(n) = \Theta\big(g(n)\big) \iff g(n) = \Theta\big(f(n)\big).$$

4. **Transpose symmetry**:
$$f(n) = \mathcal{O}\big(g(n)\big) \iff g(n) = \Omega\big(f(n)\big),$$
$$f(n) = o\big(g(n)\big) \iff g(n) = \omega\big(f(n)\big).$$

Proof. **Transitivity.** Let $0 \leq f(n) \leq c'\, g(n)$ for $n \geq n_1'$ and $0 \leq g(n) \leq c''\, h(n)$ for $n \geq n_1''$. Then $0 \leq f(n) \leq c'c''\, h(n)$ for all $n \geq \max\{n_1', n_1''\}$, and $f(n) = \mathcal{O}\big(h(n)\big)$ with $c = c'c'' > 0$. The proofs for Ω and Θ are the same, with the products of the constants. For o, let $c \in \mathbb{R}_{>0}$: the definition of $f(n) = o\big(g(n)\big)$ with the constant 1 and the definition of $g(n) = o\big(h(n)\big)$ with the constant c give two thresholds n_1' and n_1'' such that

$$0 \leq f(n) < g(n) < c\, h(n), \qquad \forall n \geq \max\{n_1', n_1''\}.$$

Since c is arbitrary, $f(n) = o\big(h(n)\big)$. The proof for ω is the same, with $0 \leq c\, h(n) < g(n) < f(n)$.

Reflexivity. Let $\tilde{n}$ be such that $f(n) \geq 0$ for all $n \geq \tilde{n}$. Then $0 \leq 1 \cdot f(n) \leq f(n) \leq 1 \cdot f(n)$ for all $n \geq \tilde{n}$: the definitions of Θ, $\mathcal{O}$ and Ω hold with all the constants equal to 1 and $n_1 = \tilde{n}$.

Symmetry. If $0 \leq c_1\, g(n) \leq f(n) \leq c_2\, g(n)$ for all $n \geq n_1$, dividing the left inequality by c_1 and the right inequality by c_2 we obtain

$$0 \leq \frac{1}{c_2}\, f(n) \leq g(n) \leq \frac{1}{c_1}\, f(n), \qquad \forall n \geq n_1,$$

that is, $g(n) = \Theta\big(f(n)\big)$. Exchanging the roles of f and g we obtain the other direction.

Transpose symmetry. We have $0 \leq f(n) \leq c\, g(n)$ if and only if $0 \leq \frac{1}{c}\, f(n) \leq g(n)$: the first inequality defines $f(n) = \mathcal{O}\big(g(n)\big)$ with the constant c, the second one defines $g(n) = \Omega\big(f(n)\big)$ with the constant $\frac{1}{c}$, and the threshold is the same. For o and ω, let $f(n) = o\big(g(n)\big)$ and let $c \in \mathbb{R}_{>0}$. The definition of $f(n) = o\big(g(n)\big)$ with the constant $\frac{1}{c}$ gives n_1 such that $0 \leq f(n) < \frac{1}{c}\, g(n)$, that is, $0 \leq c\, f(n) < g(n)$, for all $n \geq n_1$. Since c is arbitrary, $g(n) = \omega\big(f(n)\big)$. The other direction is proved in the same way. □

Because these relational properties hold for asymptotic notations, we can draw an analogy between the **asymptotic comparison** of two functions and the comparison of two real numbers a and b:

$$f(n) = o\big(g(n)\big) \quad \text{is like} \quad a < b,$$
$$f(n) = \mathcal{O}\big(g(n)\big) \quad \text{is like} \quad a \leq b,$$
$$f(n) = \Theta\big(g(n)\big) \quad \text{is like} \quad a = b,$$
$$f(n) = \Omega\big(g(n)\big) \quad \text{is like} \quad a \geq b,$$

$$f(n) = \omega\big(g(n)\big) \quad \text{is like} \quad a > b.$$

- For any two real numbers a and b, exactly one of the following three cases holds:

$$a < b, \qquad a = b, \qquad a > b \qquad \text{(trichotomy property)}.$$

- The analogous property **does not hold** for the asymptotic comparison of two functions: there exist two functions f and g such that none of the five notations holds, as the next example shows.

Example 10.16: two functions that cannot be compared

We consider the asymptotically positive function

$$f : \mathbb{N} \to \mathbb{R}, \qquad f : n \mapsto n^{1+\sin n}$$

and the function $g(n) = n$. For every $n \geq 1$ we have $\frac{f(n)}{g(n)} = n^{\sin n}$. Since π is irrational, the values $n \bmod 2\pi$, $n \in \mathbb{N}$, are dense in $[0, 2\pi]$. Hence there exist arbitrarily large integers n with $\sin n > \frac{1}{2}$, for which

$$n^{\sin n} \geq n^{1/2} = \sqrt{n},$$

and arbitrarily large integers m with $\sin m < -\frac{1}{2}$, for which

$$m^{\sin m} \leq m^{-1/2} = \frac{1}{\sqrt{m}}.$$

- To prove that

$$n^{1+\sin n} \neq \mathcal{O}(n),$$

we proceed by contradiction and we suppose that there exist $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that $0 \leq n^{1+\sin n} \leq c\,n$, that is, $n^{\sin n} \leq c$, for all $n \geq n_1$. This is false for the arbitrarily large integers n with $\sin n > \frac{1}{2}$ and $\sqrt{n} > c$.

- To prove that

$$n^{1+\sin n} \neq \Omega(n),$$

we proceed by contradiction and we suppose that there exist $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that $0 \leq c\,n \leq n^{1+\sin n}$, that is, $c \leq n^{\sin n}$, for all $n \geq n_1$. This is false for the arbitrarily large integers m with $\sin m < -\frac{1}{2}$ and $\frac{1}{\sqrt{m}} < c$.

- By Observation 10.5, it follows that

$$n^{1+\sin n} \neq \Theta(n).$$

- By the contrapositions of (10.10) and (10.12), it also follows that

$$n^{1+\sin n} \neq o(n) \qquad \text{and} \qquad n^{1+\sin n} \neq \omega(n).$$

The same conclusion follows from the limit

$$\lim_{n\to+\infty} \frac{n^{1+\sin n}}{n} = \lim_{n\to+\infty} n^{\sin n} = \lim_{n\to+\infty} e^{\sin n \,\ln n},$$

which does not exist (along the integers n with $\sin n > \frac{1}{2}$ the ratio tends to $+\infty$, along the integers m with $\sin m < -\frac{1}{2}$ it tends to 0^+): the limit is neither 0^+ nor $+\infty$, and we apply (10.9) and (10.11).

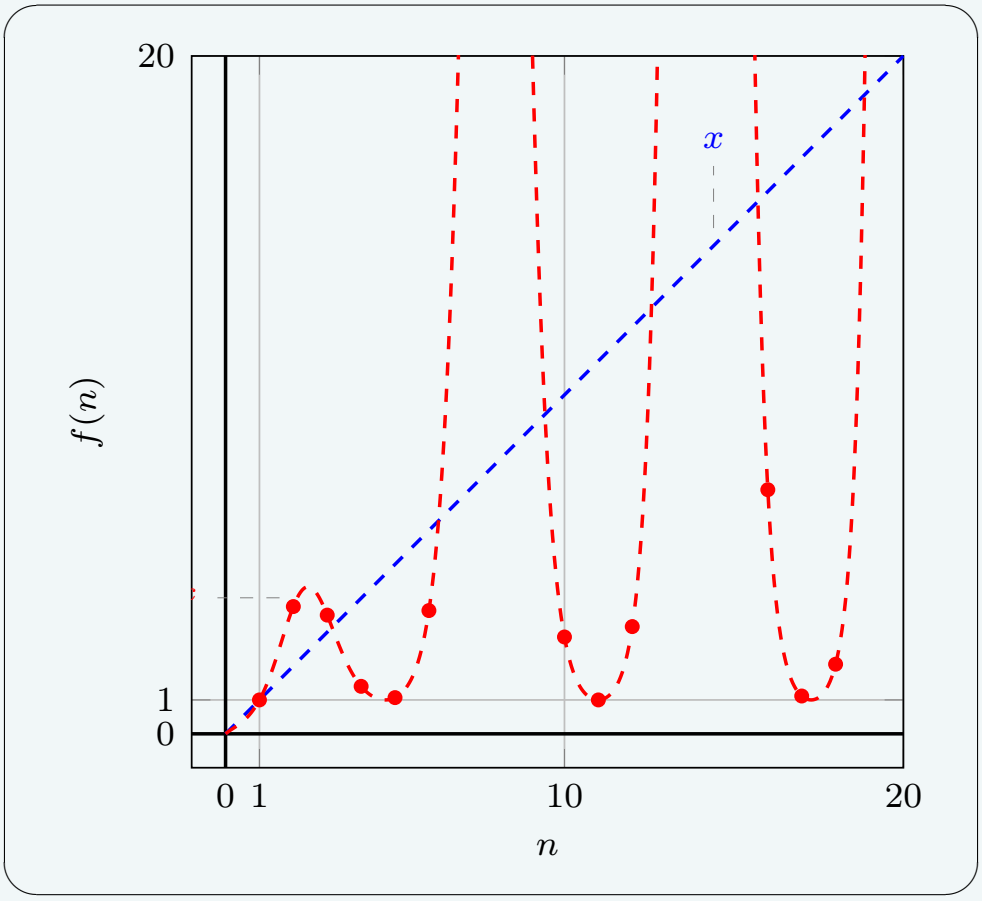
Therefore none of the five notations holds between $n^{1+\sin n}$ and n. Note that this is a stronger failure than the failure of trichotomy: not even the analogues of $a \leq b$ or $a \geq b$ hold.

Example 10.17: the graph of $n^{1+\sin n}$

This is the graph of the function

$$f : \mathbb{N} \to \mathbb{R}, \qquad f : n \mapsto n^{1+\sin n},$$

together with the graph of the function $g : \mathbb{R}_{\geq 0} \to \mathbb{R}$, $g : x \mapsto x$. The values $f(n)$ are alternately above and below the line g, farther and farther from it.



10.5 Asymptotic notation and orders of growth

In the chapter «Asymptotic growth» the orders of growth of the families of increasing functions of a natural variable were compared by computing limits of ratios. With Observation 10.11, every limit equal to 0^+ becomes a relation o.

Example 10.18: the hierarchy of the orders of growth

For every $\alpha, \beta \in \mathbb{R}_{>0}$ and $a, b \in \mathbb{R}_{>1}$, the limits

$$\lim_{n\to+\infty} \frac{\log_a^\beta n}{n^\alpha} = 0^+, \qquad \lim_{n\to+\infty} \frac{n^\alpha}{b^n} = 0^+, \qquad \lim_{n\to+\infty} \frac{b^n}{n!} = 0^+, \qquad \lim_{n\to+\infty} \frac{n!}{n^n} = 0^+$$

of the chapter «Asymptotic growth» give, by (10.9),

$$\log_a^\beta n = o(n^\alpha), \qquad n^\alpha = o(b^n), \qquad b^n = o(n!), \qquad n! = o(n^n).$$

By the transitivity of o (Proposition 10.18), each function of the chain is o of all the functions that follow it, for example $\log_a^\beta n = o(n!)$.

Given two asymptotically increasing functions f and g of a natural variable:

- if $f(n) = o\big(g(n)\big)$, f has a strictly smaller order of growth than g (f grows slower than g);
- if $f(n) = \mathcal{O}\big(g(n)\big)$, f has a smaller or equal order of growth (f grows at most as fast as g);
- if $f(n) = \Theta\big(g(n)\big)$, f has the same order of growth as g (f grows as fast as g);
- if $f(n) = \Omega\big(g(n)\big)$, f has a greater or equal order of growth (f grows at least as fast as g);
- if $f(n) = \omega\big(g(n)\big)$, f has a strictly greater order of growth than g (f grows faster than g).

10.6 Asymptotic notation in the analysis of algorithms

In the chapters «Input size» and «Running time» we measure the size $|I|$ of the input of an algorithm and we count the number of steps $S(n)$ that the algorithm executes in the worst case, as a function of a natural parameter n. The exact expression of $S(n)$ depends on the conventions used to count the steps, and the time of one step depends on the computer. The asymptotic notation keeps only what does not depend on these details: the **order of growth**.

- **Constant factors do not matter.** If $f(n) = \Theta\big(g(n)\big)$ and $\lambda \in \mathbb{R}_{>0}$, then $\lambda\, f(n) = \Theta\big(g(n)\big)$, with the constants λc_1 and λc_2 and the same threshold. For example, an algorithm executing $4n$ steps and an algorithm executing n steps both have a number of steps $\Theta(n)$.
- **Lower-order terms do not matter.** This is made precise by the next observation, which follows from Observation 10.7.

Observation 10.19: dropping the lower-order terms

Given two asymptotically non-negative functions f and g of a natural variable, we have

$$g(n) = \mathcal{O}\big(f(n)\big) \implies f(n) + g(n) = \Theta\big(f(n)\big).$$

Proof. Let $0 \le g(n) \le c\, f(n)$ for all $n \ge n_1'$, and let $f(n) \ge 0$ for all $n \ge n_1''$. Then, for all $n \ge n_1 = \max\{n_1', n_1''\}$,

$$0 \le 1 \cdot f(n) \le f(n) + g(n) \le f(n) + c\, f(n) = (1 + c)\, f(n),$$

that is, $f(n) + g(n) = \Theta\big(f(n)\big)$ with the constants $c_1 = 1$ and $c_2 = 1 + c$. □

Example 10.19: the number of steps of two algorithms

In the chapter «Running time» we count the steps of two algorithms on an array of n entries.

- `FindMaximum` executes $S(n) = 4n$ steps. By choosing $c_1 = c_2 = 4$ and $n_1 = 1$ we have $0 \le 4\,n \le S(n) \le 4\,n$ for all $n \ge 1$, so $S(n) = \Theta(n)$: the number of steps is **linear** in n.
- `InsertionSort` executes, in the worst case,

  $$S(n) = \frac{3n^2 + 9n - 8}{2}$$

  steps. To prove that $S(n) = \Theta(n^2)$ we determine $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

  $$0 \le c_1\, n^2 \le \frac{3n^2 + 9n - 8}{2} \le c_2\, n^2, \qquad \forall n \ge n_1.$$

  For the lower bound, $9n - 8 \ge 0$ for all $n \ge 1$, hence $\frac{3n^2+9n-8}{2} \ge \frac{3}{2}\, n^2$. For the upper bound, $9n - 8 \le 9n \le 9n^2$ for all $n \ge 1$, hence $\frac{3n^2+9n-8}{2} \le \frac{12\, n^2}{2} = 6\, n^2$. By choosing

$c_1 = \frac{3}{2}$, $c_2 = 6$ and $n_1 = 1$, we have

$$0 \leq \frac{3}{2}\,n^2 \leq \frac{3n^2 + 9n - 8}{2} \leq 6\,n^2, \qquad \forall n \geq 1.$$

The same result follows from the limit $\lim_{n\to+\infty} \frac{3n^2+9n-8}{2n^2} = \frac{3}{2} \in \mathbb{R}_{>0}$ and (10.13). The number of steps is **quadratic** in n.

- Since $n = o(n^2)$, for large arrays `InsertionSort` executes, in the worst case, many more steps than `FindMaximum`: by Observation 10.11, the ratio $\frac{4n}{(3n^2+9n-8)/2}$ tends to 0^+.

- The three notations $\mathcal{O}$, Ω and Θ say different things about the number of steps $S(n)$ in the worst case. If $S(n) = \mathcal{O}\big(g(n)\big)$, then, for all $n \geq n_1$, the algorithm executes at most $c\,g(n)$ steps on **every** input with parameter n. If $S(n) = \Omega\big(g(n)\big)$, then, for all $n \geq n_1$, there is **at least one** input with parameter n on which the algorithm executes at least $c\,g(n)$ steps. The statement $S(n) = \Theta\big(g(n)\big)$ gives both, and it is the most informative one.

- In the chapter «Running time» the order of growth of $S(n)$ is then expressed as a function of the input size $|I|$, using the order of growth of $|I|$ obtained in the chapter «Input size».

Summary

- $f(n) = \Theta\big(g(n)\big)$, $\mathcal{O}\big(g(n)\big)$, $\Omega\big(g(n)\big)$: there exist positive constants (c_1 and c_2, or c) and a threshold n_1 such that, for all $n \geq n_1$, $f(n)$ lies between $c_1 g(n)$ and $c_2 g(n)$, below $c\,g(n)$, above $c\,g(n)$. To prove these statements we determine the constants.

- $f(n) = o\big(g(n)\big)$, $\omega\big(g(n)\big)$: the inequality $f(n) < c\,g(n)$, or $c\,g(n) < f(n)$, holds for **every** $c > 0$ from a threshold n_1 that depends on c.

- Θ is the same as $\mathcal{O}$ and Ω together; o implies $\mathcal{O}$ and ω implies Ω. None of these implications can be reversed.

- For asymptotically positive functions, the limit of $\frac{f(n)}{g(n)}$ decides: 0^+ gives o (and $\mathcal{O}$), a number $\ell > 0$ gives Θ (and $\mathcal{O}$, Ω), $+\infty$ gives ω (and Ω). When the limit does not exist, as for $2 - \sin n$ or $2 + (-1)^n$, we use the definitions.

- The notations are transitive; Θ, $\mathcal{O}$, Ω are reflexive; Θ is symmetric; $\mathcal{O}$ and Ω, and o and ω, are transpose symmetric. Unlike real numbers, two functions may not be comparable at all, as $n^{1+\sin n}$ and n.

- In the analysis of algorithms, the notation drops constant factors and lower-order terms and keeps the order of growth of the number of steps.